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Machine Learning Techniques for Mortality Prediction in Emergency Departments: A Systematic Review

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-052663
Article Type:	Original research
Date Submitted by the Author:	24-Apr-2021
Complete List of Authors:	Naemi, Amin; University of Southern Denmark, The Maersk Mc-Kinney Moller Institute Schmidt, Thomas; University of Southern Denmark, The Maersk Mc-Kinney Moller Institute Mansourvar, Marjan; University of Southern Denmark, The Maersk Mc-Kinney Moller Institute Naghavi-Behzad, Mohammad Ebrahimi, Ali; University of Southern Denmark, The Maersk Mc-Kinney Moller Institute Wiil, Uffe ; University of Southern Denmark,
Keywords:	Health informatics < BIOTECHNOLOGY & BIOINFORMATICS, Information technology < BIOTECHNOLOGY & BIOINFORMATICS, Information management < BIOTECHNOLOGY & BIOINFORMATICS

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Machine Learning Techniques for Mortality Prediction in Emergency Departments: A Systematic Review

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Abstract

Objective: The application of machine learning (ML) methods to identify clinical deterioration remains equally challenging as identification of deterioration using track and trigger protocols and similar human-driven protocols. However, since the ML approaches are as diverse as the problem they deal with, there is a need to assess various methods and their performance. This systematic review aims to assess the performance and clinical feasibility of ML algorithms in prediction of in-hospital mortality for medical patients using vital signs at emergency departments. It is essential to analyze and merge the results of previous studies to have an updated overview of the state-of-the-art. This process provides new insights and identifies gaps to design future research.

Methods: The databases including Medline (PubMed), Scopus, and Embase (Ovid) were searched between 2010-2020, to extract published articles describing ML-based models utilizing vital signs variables to predict in-hospital mortality for medical patients at emergency departments. Based on the chosen focus area, relevant studies were retrieved, and information including study design, population, patients' outcome, vital signs types, and ML methods was extracted based on the CHARMS checklist. The risk of bias for included papers was assessed using the PROBAST tool.

Results: Ten articles were included in the final review. We found that eight ML models including logistic regression, decision tree, K-nearest neighbors, support vector machine, gradient boosting, random forest, artificial neural networks, and deep neural networks have been applied in this domain. Many studies failed to report essential main analysis steps such as data preprocessing and handling missing values. All included studies had a high risk of bias in the statistical analysis part, which could lead to poor performance in practice. Although the main aim of all studies was developing a predictive model for mortality, six articles did not provide a time horizon for the prediction.

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Conclusion: Our review provides an updated overview of the state-of-the-art and revealed research gaps; based on these, we provide eight recommendations for future studies to make the use of ML more feasible in practice. By following these recommendations, we expect to see more mature ML models applied in the future to help clinicians identify patient deterioration earlier.

Keywords: Machine learning, emergency department, mortality, prediction, healthcare, vital signs.

Strength and limitations of this study:

- To the best of our knowledge, this study is the first systematic review on the topic of using machine learning algorithms to predict mortality at emergency departments.
- The study was done with the help of an experienced research librarian, and three expert clinicians at emergency departments.
- All authors were involved in different parts of screening and analysis.
- Analysis was done using standard checklists.
- It was not possible to conduct meta-analysis.

1 INTRODUCTION

The improved life expectancy in Europe and most developed countries[1] has increased admission to the Emergency Departments (EDs), especially for elderly people who have a higher risk of morbidity and mortality[2,3]. Also, having more inter- and intra-city travels and organized emergency medical services to transfer these patients to ED brought more importance as well as patient load to EDs.[4,5] The average rate of mortality among in-patient admissions is around 2% in US,[6] indicating the necessity of using clinical deterioration prediction models.[7] Moreover, a recent study has shown that some in-hospital deaths due to clinical deterioration can be avoided.[8]

Vital signs including heart rate (HR), pulse rate (PR), respiratory rate (RR), oxygen saturation (SpO2), blood pressure (BP), body temperature (TP), and Glasgow coma scale (GCS) are essential parts of the monitoring process of patients. These data can be considered as the cheapest and most important information, which could easily be collected during patients' admission.[9] While vital signs have been used in clinical practice for over a century, few studies have effectively quantified their performance in various clinical applications.[10] In recent years, various studies have emphasized that changes in vital signs happen several hours before a serious adverse event.[11–13] Nowadays, vital signs play a key role in identification of patients at risk of deterioration in ED, however deterioration often occurs unnoticed or is not identified until it becomes too late to intervene.[9]

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Machine Learning (ML) addresses the question of how to build a computer model that automatically enhances performance through experiments, lying at the intersection of computer science, statistics, and a variety of other disciplines concerned about automatic improvement over time, and inferring and decision making under uncertainty.[14] Various ML algorithms have been developed to solve a wide variety of problems. The most widely used ML algorithms are supervised learning algorithms focusing on relationship and dependencies between the input and output features.[15] Hence, we can identify the output values for unseen data based on the relationships learned from the training datasets. Unsupervised learning is subset of ML algorithms that is used in pattern recognition and descriptive modeling without the use of output categorizes or data labeling.[16]

Because of enormous advancement in the development of modern ML algorithms such as deep learning, the availability of large databases, and increased computational power, ML has progressed rapidly over the last two decades from laboratory curiosity to practical technology in a wide range of commercial applications. Consequently, utilization of ML techniques in the healthcare domain is increasing rapidly, and the list of tasks where ML has matched or outperformed, physicians is growing. Clinical outcome prediction is one challenge receiving attention, and by improvement of ML techniques, the ML-based systems aim to outperform conventional clinical scoring systems.[14,17]

Various studies have been made in this area evaluating the prediction accuracy of in-hospital mortality, cardiac arrest, and intensive care unit (ICU) transfer. To reach this goal, various techniques such as logistic regression (LR), support vector machine (SVM), K-nearest neighbors (KNN), decision tree (DT), random forest (RF), Gaussian process (GP), and artificial neural networks (NN) have been applied, and the results confirmed the appropriateness of using ML in this area.[14] The capability of handling large datasets is the most considerable advantage of ML algorithms, which can result in more accurate prediction of clinical outcomes. Currently, we have various prediction models, using different techniques, in order to improve the triage process and to provide better identification of high-risk patients.[18,19] Although different studies made use of a variety of models, no single model has yet proved to be superior regarding the clinical efficacy.[20–25]

Despite the use of vital signs in patient monitoring and clinical decision making, the importance of specific types of vital signs, their correlation, and frequency of registration to best prevent adverse events and in-hospital mortality is still unclear.[9] However, ML could be a good solution for these challenges. Considering the possible role of ML models in predicting clinical deterioration of ED patients[26], it is essential to analyze and merge the results of previous studies to have an updated overview of the state-of-the-art. This systematic review focuses on the clinical efficacy and technical

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implications of ML-based models in prediction of mortality using vital signs predictors in medical patients admitted to EDs.

2 MATERIALS AND METHODS

This systematic review was conducted in the period from July to December 2020. The main questions addressed by this study are presented in Table 1.

2.1 Search strategy and study selection

Three databases including Medline (PubMed), Scopus, and Embase (Ovid) were targeted. Relevant articles were extracted using a broad range of relevant keywords. We stratified keywords into five groups namely, ML keywords, medical keywords, document type, publication year, and language. The keywords in a group were paired using OR operators and all groups were paired using AND operator. Table 2 shows the search keywords that were applied to the titles, abstracts, and full text in the three databases. Inclusion and exclusion criteria are listed in Table 3.

2.2 Data extraction

Three researchers (AN, MM, AE) screened the title and extracted abstracts, independently. The screening process was done using the Covidence tool. Then, two researchers (AN, TS) read the full texts that were extracted after the title and abstract screening phase to retrieve the final set of articles that met our objectives and inclusion criteria. Spreadsheets for item extraction were prepared based on the critical appraisal and data extraction for systematic reviews of prediction modelling studies (CHARMS) checklist.[27] The extracted items include study design, publication date, source of data, study population, outcomes, time horizon, list of vital signs predictors, data preprocessing, model development, model performance, model validation and evaluation, and interpretation.

2.3 Risk of bias assessment

Risk of bias (ROB) for each study was assessed by using the prediction risk of bias assessment tool (PROBAST) checklist[28] and reported based on an adapted form.[29] Extracted articles were classified into three different categories (low, high, unclear). A study is considered as having a high ROB if the study has at least one high ROB in four domains. ROB assessment of each study was checked by two of researchers (AN, MNB); disagreements were resolved in collaboration with a senior researcher (UKW).

2.4 Patient and public involvement

No patient involved.

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3 RESULTS

The initial search resulted in 6,899 records, which after removing duplicates, evaluation of eligibility for inclusion, full-text assessment, quality assessment and review of the references of the included papers, was narrowed down to ten articles. The study selection flowchart is shown in Figure 1.

General information, as well as study design and population of included studies, are summarized in Table 4. Six out of ten articles were published after 2019, which indicates an increased interest recently among researchers to explore the capability of ML in prediction of clinical outcomes in EDs.

3.1 Studies design

Among extracted articles, nine studies have used electronic data of the related hospitals, while one of the studies was based on registry-based public data.[30] One of the essential parts in utilizing ML techniques in each application is gathering and preparation of the dataset, which has a high impact on the performance of the final ML-based system. In general, ML techniques need an extensive dataset to produce good results, and the performance of ML models depends largely on the number of training examples.[14] Based on Riley et al.,[31] nine studies had a high enough number of patients to build acceptable models through their datasets, while one study proposed a model based on only 100 patients.[28]

3.2 Vital signs predictors

We considered studies, which utilized vital signs among their predictors in building models. The vital signs set used in extracted articles included HR, PR, RR, SpO2, BP, TP, and GCS (Table 4). BP and TP were the most widely used vital sign predictors among the included studies. SpO2, RR, and HR were common vital signs, which were used in eight, seven, and seven articles, respectively, while GCS and PR were the least commonly used vital signs. Besides the vital signs, all studies used other clinical predictors such as demographic variables, blood tests, comorbidities, diagnoses, and triage information.

3.3 Data preparation

Real-world datasets often need to be transformed, cleaned or changed before use. These datasets may contain missing values, noise, and even wrong entries or inconsistencies.[32] Missing values is one of the main challenges in the clinical data mining[33] that often provide additional information about patients. Therefore, in almost all cases, raw data will need to be preprocessed before it can be used as basis for ML modeling. This preprocessing may affect the final model, and the specific steps are important to clarity. However, six articles did not utilize proper techniques to impute missing values, and excluded up to 32% of incomplete patients' records in some studies.[22,30] Different approaches, including

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replacing missing values with median,[34] considering missing values as a categorical variable,[35] stochastic single imputation technique,[20] and Gaussian process technique[36] were applied by the remaining four articles to impute missing values.

3.4 Outcome and time horizon

The percentage of target patients in the included studies, except,[37] varies between 0.40-5.33%. It should be noted that, Liu et al.[37] have conducted their study on a small dataset containing 100 patients from which 40 patients died during hospitalization. Since the outcome patients in most studies were rare (<6%), large datasets should be collected to obtain enough event patients.

As the main aim of the included studies was building ML models to predict mortality at EDs, these articles should specify how much earlier their models can predict mortality. However, as shown in Table 5, only four studies reported their time horizon for mortality prediction. 9-12 minutes, 2 hours, 24 hours, 72 hours, and 28 days ahead were reported as time horizons. Since studies in this area try to predict mortality in advance, it is necessary that all studies report their time horizons, meaning the time that their models provide before mortality occurs.

3.5 Imbalanced data

Nine studies utilized classification approaches to predict mortality at EDs. Since the number of patients who died during hospitalization (Table 4) for all studies, except,[37] is less than 6%, some proper preprocessing techniques such as balancing classes is needed. In addition, except,[36] all other studies stratified patients into two groups, e.g. dead patients and survived patients, but the number of patients who died during hospitalization is much lower. This challenge, which is called imbalanced data, can affect the performance of ML model significantly; thus, proper techniques should be applied to address this challenge. Nevertheless, only one study[23] mentioned this challenge and used Synthetic Minority Over-sampling Technique (SMOTE) to address it.[38]

3.6 Machine Learning models

As shown in Table 5, eight ML techniques have been utilized in the included studies. RF was the most common ML technique and was used in six studies. A complete list of advantages and disadvantages of each ML technique is summarized in Table 6. Although most studies have been made during the last five years, we observed traditional ML algorithms such as LR more than advanced algorithms, applied to build predictive models in four studies.

3.7 Validation and evaluation

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Six articles used k-fold cross-validation,[22,23,30,34,35,37] while one study applied bootstrapping technique for validation purpose.[20] Three studies split their data into training and test sets and used a portion of the training set for validation.[18,19,36] Three studies investigated the generalization capability of their models using external validation, while the rest of the studies were internally validated. Researchers have used different metrics for evaluation of ML models in predicting mortality at EDs. The area under the curve (AUC) metric was used in nine articles. Three studies assessed the performance of ML models using sensitivity and specificity. The Brier Score was used in three studies as evaluation metric, while two studies presented accuracy of models.

3.8 Personalized model

Patients' characteristics should be considered in developing predictive models and its impact on the improvement of models' performance should be investigated. However, among the included studies, only one study used information of each patient to predict the risk of mortality for the same patient.[36] In that study, researchers introduced a ML model to follow up on the trends of vital signs of each patient and by that they tried to consider patients' characteristics, while in the other studies, the researchers built binary classifiers using different ML algorithms for the whole population.

3.9 Risk of bias assessment

There is a significant amount of incomplete reporting within the studies' results. For stance, most studies did not report information about the tuning process of ML models and hyperparameters values considered during the model training process.[18–20,22,23,30,34,35,37] A summary of the PROBAST assessment is shown in Table 7. As shown, participants' selection was done properly except in one study.[37] However, most studies had a high ROB in the sample size, handling missing data, and statistical analysis domain.

4 DISCUSSION

4.1 Limitations of included studies

This review consisted of ten studies that developed ML-based models to predict mortality in EDs. We found many methodological and reporting shortcomings, and this can lead to poorer performance than reported when the models are deployed in real life. Many studies skipped or failed to report important phases such as data preprocessing, handling missing values, and model development. It has been stated that up to 80% of data analysis is spent on data cleaning and data preparation,[32] so these phases are very crucial for having a ML predictive model with good performance. It also could be one of the reasons why some researchers have discussed poor effectiveness of ML predictive models in clinical environments.[39,40] Moreover, despite the dramatic advances in ML and introducing novel and effective models,

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various researchers still have applied traditional ML techniques such as LR and DT. For instance, we found that LR is the second most common algorithm used in the studies. Most ML models have parameters that needed to be tuned to gain the best performance and generalization power and avoiding overfitting, but most of studies did not provide information about the models' validation process and finding the hyperparameters of the models. Of ten studies, nine studies are retrospective studies and only three studies have evaluated their models based on external validation.

The ultimate aim of included studies is to predict mortality in EDs ahead of time to provide enough time for clinicians to intervene and prevent mortality. However, only four studies have provided their time horizon, the time that clinicians have to treat the patients. The other studies have built binary classification algorithms to stratify patients in two classes, and their models do not have the capability to be used in practice.

4.2 Clinical considerations and prospective

A recent systematic review on the value of vital sign trends in monitoring and predicting clinical deterioration showed lack of research and knowledge regarding the importance of vital signs in clinical deterioration of ED patients.[9] Even though considering the vital sign as cheap and available clinical predictors, there is a hypothesis stating the lack of evidence to support the usefulness of continuous monitoring of vital signs as a daily routine of clinical practice.[41–43] The other barrier refers to comparison between ML methods and conventional logistic regression for clinical prediction models. A recent systematic review showed no performance benefits of ML methods over logistic regression.[44] Therefore, further studies need to be performed to prove superiority of ML algorithms over conventional models. It seems that there is still a long way to go for having ML algorithms as the choice of clinical deterioration in EDs. The increase of studies in this field during the last few years, demonstrates an increased attention to appraise the capabilities of ML methods in clinical practice.

4.3 Challenges and recommendations for future work

Perhaps in this research area, the main challenge is integration of ML models into clinical practice. Despite the great research successes in building ML-based predictive models for clinical practice, there remain few examples of ML models being successfully integrated into the daily routine or critical parts of clinical environments.[45] This reveals that what is being done in research is not completely in line with the realities of the clinical practice. In this section, we present recommendations on how to address the most prominent issues and encourage researchers to take them into account when building ML-based predictive models for EDs.

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Our analysis indicated that most studies are retrospective. However, the impact of ML techniques in clinical environment will need further validation in randomized control trials and prospective studies before widespread clinical adoption where the data could be incomplete and noisy with high level of uncertainty. *Recommendation 1*: we recommend that instead of building models using historical collected data, researchers consider prospective analysis and think about building adaptive models that can monitor patients in real-time. *Recommendation 2*: we recommend that researchers investigate the performance of their models using external validation to have a better estimation of the generalization power of their models. *Recommendation 3*: regarding the preprocessing phase, since some steps such as data cleaning, handling missing and noisy values are an inevitable part of each data analysis study, we recommend that researchers instead of ignoring records with missing values, use proper techniques to handle noisy and incomplete records. For instance, we recommend that complete case analysis is avoided and proper imputation techniques such as iterative multivariate imputation technique is used instead;[46] this is one of the best approaches for imputing missing values. Another challenge in most studies is that the number of event patients is too low (<6%) compared to non-event patients. This can lead to bias and overfitting during developing ML models, so proper under/oversampling techniques such as SMOTE and its variations should be utilized to make a balance between classes before developing models. Another issue is time horizon for prediction. Gerry et al. recommended that time horizon should be limited to a few days at most, since signs of deterioration will probably not be observed for more than a few days.[47] *Recommendation 4*: we suggest that researchers in this area choose a realistic time horizon rather than prediction of mortality after six or twelve months. Moreover, most of the included studies are based on building a single model for the whole population. *Recommendation 5*: as patients have different characteristics and norms, we recommend that patients' characteristics, vital signs trajectories, trends, etc. are used for each individual patient to tune ML models to have a better performance in practice. *Recommendation 6*: we recommend that researchers find a way to make their results and workflow understandable and interpretable by humans. For instance, deep learning is one of the hottest research topics today, and it has been applied in different applications of clinical practice, but it has inherent limitations. For example, despite impressive results, it is not indicative of high-level reasoning and rationality. Such methods are often considered as a "black box", where the decision logic is presented in millions of numerical weights and biases. This lack of transparency could have some legal and ethical implications and might increase distrust of ML models by patients and clinicians, and it might be one of the reasons that patients are likely to trust a clinician more than a machine. This has been described as a crucial problem for ML acceptance in clinical practice and attempts are underway to create more

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human interpretable models. For example, with the help of image processing techniques, it is possible now to visualize the input and output of ML models; this makes the understanding of ML models easier. [17]

Lack of widely accepted and utilized reporting or publication guidelines for implementing ML in clinical practice is challenge that brings difficulty to the research quality assessment, especially for clinicians without a strong mathematics and computer science background.[17] Another challenge for ML models is that they need huge amount of labeled data to be trained, and the performance of ML models often depends on the performance of the human labelling the data. *Recommendation 7*: as this area is interdisciplinary in nature, it is imperative that clinicians and artificial intelligence experts become better at interdisciplinary collaboration to become more familiar with the needs, limitations, and challenges in clinical and technical domains, which ultimately increases the quality of the predictive models. *Recommendation 8*: we recommend that researchers should be careful and accurate when reporting their studies. It has been shown that the quality of reporting of predictive models in this clinical domain is poor. Only with full and transparent reporting of information about all aspects of a predictive model, the risk of bias and the usefulness of that model can be assessed. It is recommended that researchers use available checklists such as SPIRIT-AI[48] for designing the trial, and TRIPOD[49] and CONSORT-AI[50] for reporting the findings to make sure that they provide sufficient information to clearly and accurately report their studies.

5 CONSLUSION

To the best of our knowledge, this study is the first review on the topic of utilization of ML techniques to predict mortality at EDs using vital signs. Our systematic review of the literature provides an updated overview of the state-of-the-art on this topic (covering 2010 to 2020). Initially, 6,899 potential articles were identified of which ten were included in the analysis. After a comprehensive review of these ten articles, we generated eight recommendations to increase the feasibility of implementing ML models in EDs. These recommendations provide actionable suggestions to be used to increase the quality of future work in this area.

Acknowledgement

The authors would like to thank Dr. Mette Brandt Eriksen, research librarian at University of Southern Denmark library for help with the search strategy and compiling the literature databases, and Prof. Annmarie Touborg Lassen, Prof. John Kellet, and Prof. Mikkel Brabrand, expert emergency department clinicians at Odense University Hospital, for their help during performing this systematic review.

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Author contributions: All authors were involved in conception and design of the study. Literature search was conducted by AN. AN, TS, MM, and AE participated in screening and extracting relevant studies. AN, MNB, TS, and UKW were responsible for analyzing, and drafting the manuscript. All authors made critical revisions. AN takes responsibility for the paper as a whole.

Funding: There is no funding attached to this systematic review.

Competing interests: None declared.

Patient consent for publication: Not required.

Data availability statement: All data relevant to the study are included in the article or uploaded as online supplementary information.

Ethics statement: Formal ethical approval is not required, as primary data will not be collected. The review will be disseminated in peer-reviewed publications.

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Figure 1. Flow diagram of study selection (PRISMA chart).

Table 1. Research questions	
Q1	What ML techniques have been used to predict in-hospital mortality in EDs?
Q2	What are the common vital signs variables used in studies?
Q3	How researchers prepared data for ML techniques?
Q4	What are the approaches to solve the problem (e.g. binary classification or time series prediction)?
Q5	What are the challenges and open issues in this domain?

Table 2. Search keywords in different groups	
Group 1 – ML keywords	Artificial intelligence, machine learning, deep learning, learning algorithms, supervised machine learning, unsupervised machine learning
Group 2 – Medical keywords	Clinical deterioration, mortality, in-hospital mortality, death, vital sign, emergency departments
Group 3 – Document type	Journal
Group 4 – Publication year	1st January 2010 to 1st October 2020
Group 5 – Language	English
Final result	(Group 1) AND (Group 2) AND (Group 3) AND (Group 4) AND (Group 5)

Table 3. Inclusion and exclusion criteria	
Inclusion Criteria	**Exclusion criteria**
Prediction of in-hospital mortality prediction should be the main aim of the study.	Studies which are conference articles, posters, abstracts, books or book chapters, and review articles.
The study should be done at ED.	Studies which were done in pediatrics field.
The study should be done on adult patients.	Non-English studies
ML algorithms should be used for the prediction task.	
Vital signs variables should be among the predictors used to build ML models.	

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Table 4. General characteristics of included studies

ID	Authors	Year	Country	Study type	Population	Outcome portion (%)	Patients type	Mean age (y)	Male (%)	Vital signs
A1[22]	Faisal et al.	2020	UK	Retrospective	24,696	5.33	All admissions	63	46.9	RR, SpO2, SBP, PR, GCS, TP
A2[30]	Raita et al	2019	USA	Retrospective	135,470	2.10	All admissions	46	56.8	TP, PR, SBP, DBP, RR, SpO2
A3[19]	Klug et al.	2020	Israel	Retrospective	799,522	1.55	All admissions	55	51.5	TP, HR, SBP, DBP, SpO2
A4[23]	Kwon et al.	2020	South Korea	Retrospective	23,587	3.98	Infectious patients	63	46.1	SBP, RR, TP, HR
A5[37]	Liu et al.	2011	Singapore	Retrospective	100	40.00	All admissions	65	63	RR, PR, SBP, DBP, SpO2, GCS
A6[35]	Taylor et al.	2015	USA	Retrospective	5,278	4.92	Infectious patients	65	55	HR, TP, SBP, DBP, RR, SpO2
A7[18]	Levin et al.	2017	USA	Retrospective	172,726	0.40	All admissions	46	44.7	TP, HR, RR, SBP, SpO2
A8[20]	Coslovsky et al.	2015	Switzerland	Prospective	7,680	4.60	All admissions	58	62	SBP, HR, SpO2, TP, GCS
A9[34]	Perng et al.	2019	Taiwan	Retrospective	42,220,	4.71	Infectious patients	64	56.5	SBP, GCS, TP, HR
A10[36]	Shamout et al.	2020	UK	Retrospective	37,284	0.80	All admissions	68	48.8	HR, SBP, RR, TP, SpO2

HR: heart rate; PR: pulse rate; RR: respiratory rate; SpO2: oxygen saturation; SBP: systolic blood pressure; DBP: diastolic blood pressure; TP: body temperature: GCS: glasgow coma scale

Table 5. Selected articles list and their ML-related characteristics

Id	ML algorithms	Evaluation Metrics	Time Horizon	Personalized	Handling missing values	Hyperparameter Optimization	Approach	Validation
A1	LR, RF, SVM, NN	AUC, Brier Score	-	No	No	Yes	Binary classification	External
A2	LR, RF, GB DT, DNN	AUC, Sensitivity, Specificity	-	No	No	Yes	Binary classification	Internal
A3	GB	AUC, Sensitivity, Specificity, Brier Score	-	No	No	No	Binary classification	Internal
A4	GB, RF	AUC	72h	No	No	Yes	Binary classification	External
A5	NN, SVM	Accuracy, Sensitivity, Specificity	9-12 minutes	No	No	No	Binary classification	Internal
A6	RF, DT, LR	AUC	-	No	Yes	No	Binary classification	Internal
A7	RF	AUC	-	No	No	No	Binary classification	Internal
A8	LR	AUC, Brier Score	-	No	Yes	No	Binary classification	Internal
A9	KNN, SVM, RF, DNN	AUC, Accuracy	72h, 28 days	No	Yes	No	Binary classification	Internal
A10	DNN	AUC	2h, 24h	Yes	Yes	No	Time series regression	External

ML: machine learning, LR: logistic regression, SVM: support vector machine, KNN: K-Nearest neighbors, DT: decision tree, RF: random forest, GP: gaussian process, NN: neural networks, AUC: area under the curve

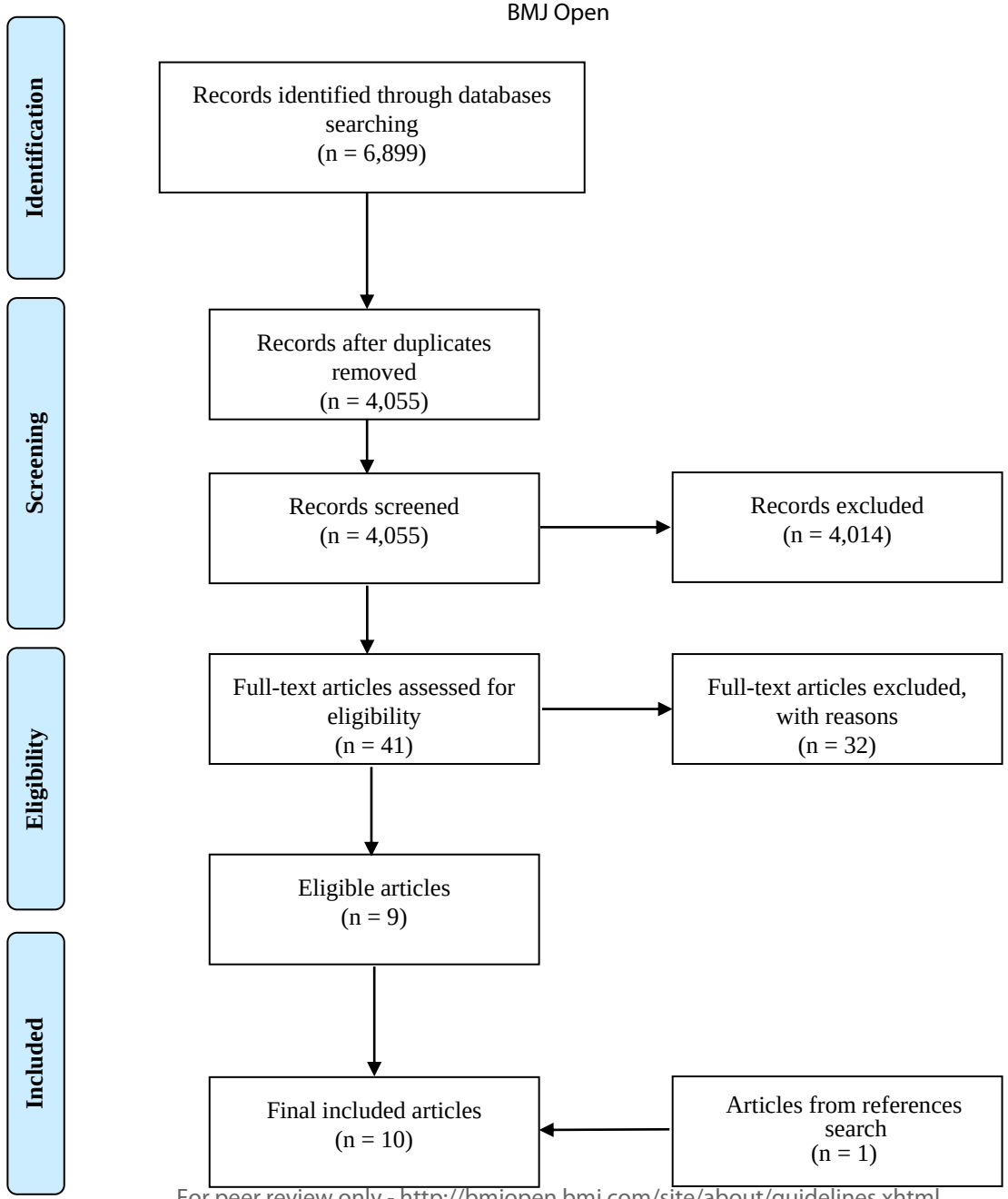
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Table 6. Summary of Machine Learning algorithms' description, advantages and disadvantages			
Algorithm	**Description**	**Advantages**	**Disadvantages**
LR[51]	LR is a supervised ML algorithm adopted from linear regression. It can be used for classification problems and finding the probability of an event happening.	Fast training, good for small datasets, easy to understand.	Not very accurate, not proper for nonlinear problems, high chance of overfitting, not flexible to adopt to complex datasets.
DT[52]	DT is a supervised ML algorithm that solves a problem by transforming the data into a tree representation where each internal node represents an attribute and each leaf denotes a class label.	Easy to understand and interpret, robust to outliers, no standardization or normalization required.	High chance of overfitting, not suitable for large datasets, adding new samples lead to regeneration of the whole tree.
KNN[53]	KNN is a supervised and instance-based ML algorithm. It can be used when we want to forecast a label of a new sample based on the similar samples with known labels. Different similarity or distance measures such as Euclidean can be used.	Simple and easy to understand.	Memory intensive, costly, slow performance, all training data might be involved in decision making.
SVM[54]	SVM is a supervised and instance-based ML algorithm that creates a boundary called hyperplane between classes. The primary aim of this algorithm is to maximize the margin between classes.	Efficient in high dimensional spaces. Effective when the number of dimensions is greater than the number of samples, long training time.	Not suitable for large datasets, not suitable for noisy datasets, regularization capabilities which prevent overfitting, handling non-linear data.
GB[55]	GB is a supervised ML algorithm, which produces a model in the form of an ensemble of weak prediction models.	High accuracy, high flexibility, robust to missing values and overfitting.	Computationally expensive, not suitable for small datasets, many parameters to optimize.
RF[56]	RF is an ensemble and supervised ML algorithm that is based on the bagging technique, meaning several subsets of data are chosen randomly with replacement and each subset is used to train one model such as DT. The output is the average of all predictions from different single models.	High accuracy, fast execution, useful for regression and classification, robust to missing values and overfitting.	Not suitable for limited datasets.
NN[57]	NN is a family of supervised ML algorithms. It is inspired by biological neural network of the human brain. NN consists of input, hidden, output layers and multiple neurons (nodes) carry data from input layer to output layer.	Accurate, suitable for complex, non-linear classification, and regression problems.	Very slow to train and test, requires large amount of data, computationally expensive and prone to overfitting.
DNN[58]	DNN is a family of supervised ML algorithms. DNN is based on NN where the adjective "deep" comes from the use of multiple layers in the network. It provides the opportunity to solve complex problems when the data is very diverse, unstructured, and inter-connected.	High accuracy, features are automatically deduced and optimally tuned, robust to noise, architecture is flexible.	Needs very large amount of data, computationally expensive, not easy to understand, no standard theory in selecting the right settings, difficult for less skilled researchers.
LR: Logistic Regression, DT: Decision Tree, KNN: K-Nearest Neighbors, SVM: Support Vector Machine, GB: Gradient Boosting, RF: Random Forest, NN: Neural Networks, DNN: Deep Neural Networks.			

Table 7. Risk of bias (ROB) assessment of included studies according to PROBAST checklist					
ID	**PROBAST items**				
	Participants	**Predictors**	**Outcomes**	**Sample size & missing data**	**Statistical analysis**
A1	+	+	+	-	-
A2	+	+	+	-	-
A3	+	?	+	-	-
A4	+	?	+	-	-
A5	-	?	+	-	-
A6	+	+	?	+	?
A7	+	+	+	-	-
A8	+	?	?	+	-
A9	+	?	?	-	?
A10	+	+	+	+	?

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Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Machine Learning Techniques for Mortality Prediction in Emergency Departments: A Systematic Review	Page 1
ABSTRACT			
Abstract	2	**Objective**: The application of machine learning (ML) methods to identify clinical deterioration remains equally challenging as identification of deterioration using track and trigger protocols and similar human-driven protocols. However, since the ML approaches are as diverse as the problem they deal with, there is a need to assess various methods and their performance. This systematic review aims to assess the performance and clinical feasibility of ML algorithms in prediction of in-hospital mortality for medical patients using vital signs at emergency departments. It is essential to analyze and merge the results of previous studies to have an updated overview of the state-of-the-art. This process provides new insights and identifies gaps to design future research. **Methods**: The databases including Medline (PubMed), Scopus, and Embase (Ovid) were searched between 2010-2020, to extract published articles describing ML-based models utilizing vital signs variables to predict in-hospital mortality for medical patients at emergency departments. Based on the chosen focus area, relevant studies were retrieved, and information including study design, population, patients' outcome, vital signs types, and ML methods was extracted based on the CHARMS checklist. The risk of bias for included papers was assessed using the PROBAST tool. **Results**: Ten articles were included in the final review. We found that eight ML models including logistic regression, decision tree, K-nearest neighbors, support vector machine, gradient boosting, random forest, artificial neural networks, and deep neural networks have been applied in this domain. Many studies failed to report essential main analysis steps such as data preprocessing and handling missing values. All included studies had a high risk of bias in the statistical analysis part, which could lead to poor performance in practice. Although the main aim of all studies was developing a predictive model for mortality, six articles did not provide a time horizon for the prediction. **Conclusion**: Our review provides an updated overview of the state-of-the-art and revealed research gaps; based on these, we provide eight recommendations for future studies to make the use of ML more feasible in practice. By following these recommendations, we expect to see more mature ML models applied in the future to help clinicians identify patient deterioration earlier.	Page 1, 2
INTRODUCTION			
Rationale	3	The improved life expectancy in Europe and most developed countries has increased admission to the Emergency Departments (EDs), especially for elderly people who have a higher risk of morbidity and mortality. Also, having more inter- and intra-city travels and organized emergency medical services to transfer these patients to ED brought more importance as well as patient load to EDs. The average rate of mortality among in-patient admissions is around 2% in US, indicating the necessity of using clinical deterioration prediction models. Moreover, a recent study has shown that some in-hospital deaths due to clinical deterioration can be avoided. Vital signs including heart rate (HR), pulse rate (PR), respiratory rate (RR), oxygen saturation (SpO2), blood pressure (BP), body temperature (TP), and Glasgow coma scale (GCS) are essential parts of the monitoring process of patients. These data can be considered as the cheapest and most important information, which could easily be collected during patients' admission. While vital signs have been used in clinical practice for over a century, few studies have effectively quantified their performance in various clinical applications. In recent years, various studies have emphasized that changes in vital signs happen several hours before a serious adverse event. Nowadays, vital signs play a key role in identification of patients at risk of deterioration in ED, however deterioration often occurs unnoticed or is not identified until it becomes too late to intervene. Machine Learning (ML) addresses the question of how to build a computer model that automatically enhances performance through experiments, lying at the intersection of computer science, statistics, and a variety of other disciplines concerned about automatic improvement over time, and inferring and decision making under uncertainty. Various ML algorithms have been developed to solve a wide variety of problems. The most widely used ML algorithms are supervised learning algorithms focusing on relationship and dependencies between the input and output features. Hence, we can identify the output values for unseen data based on the relationships learned from the training datasets. Unsupervised learning is subset of ML algorithms that is used in pattern recognition and descriptive modeling without the use of output	Page 2, 3

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
		categorizes or data labeling. Because of enormous advancement in the development of modern ML algorithms such as deep learning, the availability of large databases, and increased computational power, ML has progressed rapidly over the last two decades from laboratory curiosity to practical technology in a wide range of commercial applications. Consequently, utilization of ML techniques in the healthcare domain is increasing rapidly, and the list of tasks where ML has matched or outperformed, physicians is growing. Clinical outcome prediction is one challenge receiving attention, and by improvement of ML techniques, the ML-based systems aim to outperform conventional clinical scoring systems. Various studies have been made in this area evaluating the prediction accuracy of in-hospital mortality, cardiac arrest, and intensive care unit (ICU) transfer. To reach this goal, various techniques such as logistic regression (LR), support vector machine (SVM), K-nearest neighbors (KNN), decision tree (DT), random forest (RF), Gaussian process (GP), and artificial neural networks (NN) have been applied, and the results confirmed the appropriateness of using ML in this area. The capability of handling large datasets is the most considerable advantage of ML algorithms, which can result in more accurate prediction of clinical outcomes. Currently, we have various prediction models, using different techniques, in order to improve the triage process and to provide better identification of high-risk patients. Although different studies made use of a variety of models, no single model has yet proved to be superior regarding the clinical efficacy.	
Objectives	4	Despite the use of vital signs in patient monitoring and clinical decision making, the importance of specific types of vital signs, their correlation, and frequency of registration to best prevent adverse events and in-hospital mortality is still unclear. However, ML could be a good solution for these challenges. Considering the possible role of ML models in predicting clinical deterioration of ED patients, it is essential to analyze and merge the results of previous studies to have an updated overview of the state-of-the-art. This systematic review focuses on the clinical efficacy and technical implications of ML-based models in prediction of mortality using vital signs predictors in medical patients admitted to EDs.	Page 3
METHODS			
Eligibility criteria	5	**Inclusion Criteria:** 1 - Prediction of in-hospital mortality prediction should be the main aim of the study. 2- The study should be done at ED. 3- The study should be done on adult patients. 4- ML algorithms should be used for the prediction task. 5- Vital signs variables should be among the predictors used to build ML models. **Exclusion Criteria**: 1- Studies which are conference articles, posters, abstracts, books or book chapters, and review articles. 2- Studies which were done in pediatrics field. 3- Non-English studies	14 (table 3)
Information sources	6	Medline (PubMed), Scopus, and Embase (Ovid)	4
Search strategy	7	Scopus: (TITLE-ABS-KEY ((clinical W/2 deteriorat*) OR "mortalit*" or "death" or "hospital mortality" or "vital sign*" or "emergency department") AND TITLE-ABS-KEY ("artificial intelligenc*" or "machine learn*" or "artificial neural network*" or "bayesian learn*" or "classification algorithm*" or "deep learn*" or "fuzzy system*" or "hidden markov model*" or "k nearest neighbor*" or "learning algorithm*" or "supervised machine learn*" or "support vector machine*" or "unsupervised machine learn*")) **Embase**: ("Deterioration/" or "hospital mortality/" or "vital signs/" or ((clinical adj3 deteriorat*) or "hospital mortality" or "vital sign*").mp. [mp=ti, ab, hw, tn, ot, dm, mf, dv, kw, fx, dq, nm, kf, ox, px, rx, ui, sy]) and ("artificial intelligence/" or "machine learning" or "artificial neural network/" or "Bayesian learning/" or "classification algorithm/" or "deep learning/" or "fuzzy system/" or "hidden markov model/" or "k nearest neighbor/" or "learning algorithm/" or "supervised machine learning/" or "support vector machine/" or "unsupervised machine learning/" or ("artificial intelligenc*" or "machine learn*" or "artificial neural network*" or "Bayesian learn*" or "classification algorithm*" or "deep learn*" or "fuzzy system*" or "hidden makov model*" or "k nearest neighbor*" or "learning algorithm*" or "supervised machine learn*" or "support vector machin*" or "unsupervised	Page 14 (table 2)

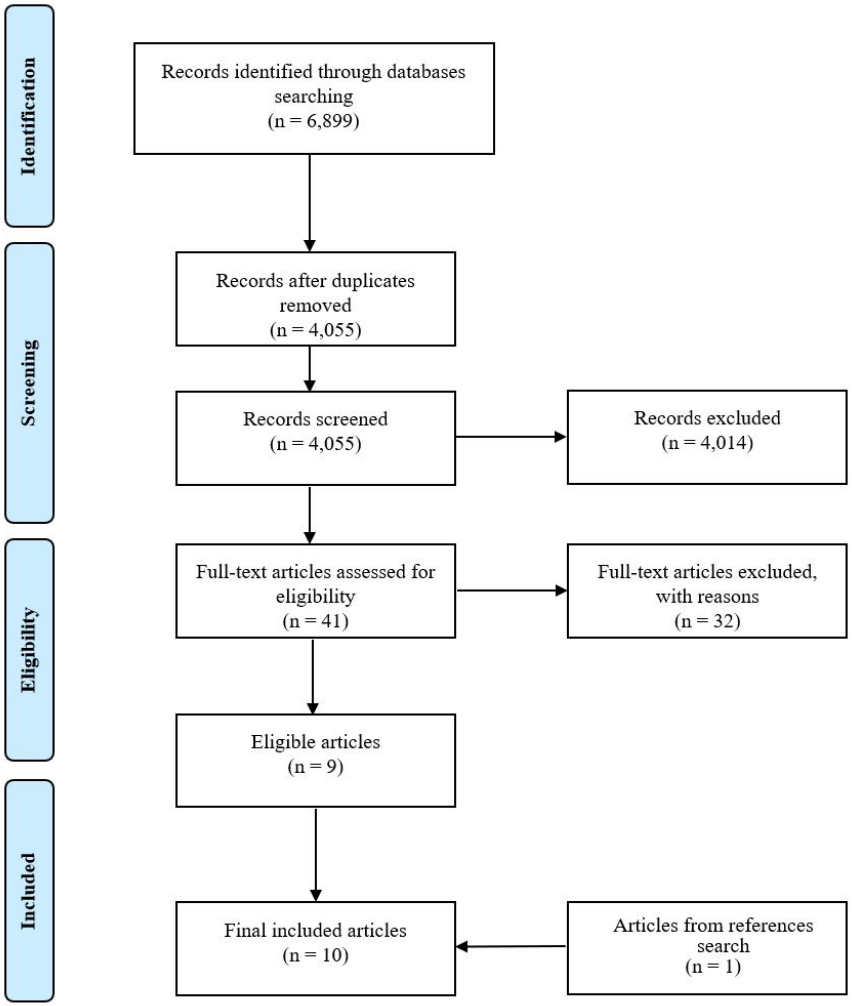
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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
		machine learn*”).mp. [mp=ti, ab, hw, tn, ot, dm, mf, dv, kw, fx, dq, nm, kf, ox, px, rx, ui, sy]) **Medline**: (“Deterioration/” or “hospital mortality/” or “vital signs/” or ((clinical adj3 deteriorat*) or “hospital mortality” or “vital sign*” or “emergency department”).mp. [mp=ti, ab, hw, tn, ot, dm, mf, dv, kw, fx, dq, nm, kf, ox, px, rx, ui, sy]) and (“artificial intelligence/” or “machine learning” or “artificial neural network/” or “Bayesian learning/” or “classification algorithm/” or “deep learning/” or “fuzzy system/” or “hidden markov model/” or “k nearest neighbor/” or “learning algorithm/” or “supervised machine learning/” or “support vector machine/” or “unsupervised machine learning/” or (“artificial intelligenc*” or “machine learn*” or “artificial neural network*” or “Bayesian learn*” or “classification algorithm*” or “deep learn*” or “fuzzy system*” or “hidden makov model*” or “k nearest neighbor*” or “learning algorithm*” or “supervised machine learn*” or “support vector machin*” or “unsupervised machine learn*”).mp. [mp=ti, ab, hw, tn, ot, dm, mf, dv, kw, fx, dq, nm, kf, ox, px, rx, ui, sy])	
Selection process	8	Three databases including Medline (PubMed), Scopus, and Embase (Ovid) were targeted. Relevant articles were extracted using a broad range of relevant keywords. We stratified keywords into five groups namely, ML keywords, medical keywords, document type, publication year, and language. The keywords in a group were paired using OR operators and all groups were paired using AND operator. Table 2 shows the search keywords that were applied to the titles, abstracts, and full text in the three databases	Page 4
Data collection process	9	For item extraction were prepared based on the critical appraisal and data extraction for systematic reviews of prediction modelling studies (CHARMS) checklist. The extracted items include study design, publication date, source of data, study population, outcomes, time horizon, list of vital signs predictors, data preprocessing, model development, model performance, model validation and evaluation, and interpretation.	Page 4
Data items	10a	Mortality is the outcome.	Page 14 (Table 1)
	10b	Study year, country, study type (retrospective, prospective), population, outcome portion (%), patients type, mean age, male (%), vital signs, machine learning algorithms, evaluation metrics, time horizon for prediction, personalized model, handling missing values, hyperparameter optimization, approach for prediction, validation	Page 15 (table 4, 5)
Study risk of bias assessment	11	Risk of bias (ROB) for each study was assessed by using the prediction risk of bias assessment tool (PROBAST) checklist and reported based on an adapted form. Extracted articles were classified into three different categories (low, high, unclear). A study is considered as having a high ROB if the study has at least one high ROB in four domains. ROB assessment of each study was checked by two of researchers (AN, MNB); *disagreements were resolved in collaboration with a senior researcher* (UKW).	Page 4
Effect measures	12	It is nor relevant to this study.	-
Synthesis methods	13a	Three researchers (AN, MM, AE) screened the title and extracted abstracts, independently. The screening process was done using the Covidence tool. Then, two researchers (AN, TS) read the full texts that were extracted after the title and abstract screening phase to retrieve the final set of articles that met our objectives and inclusion criteria. Spreadsheets	Page 4
	13b	-	
	13c	PRISMA flowchart was used to visualize the process of selecting the final set of studies. Also extracted information of studies are presented by two tables (table 4 and table 5).	Figure 1, table 4,5
	13d	-	
	13e	-	
	13f	-	
Reporting bias assessment	14	(PROBAST) checklist is used in this study.	Page 15 (table 7)

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
Certainty assessment	15	The outcome is obvious (in-hospital mortality).	Page 14 (Table 1)
RESULTS			
Study selection	16a		Figure 1
	16b	-	
Study characteristics	17	**Table 4. General characteristics of included studies** ID | Authors | Year | Country | Study type | Population | Outcome | Patients type | Mean | Male | Vital signs	Page 15, 16

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported

						portion (%)		age (y)	(%)	
A1[22]	Faisal et al.	2020	UK	Retrospective	24,696	5.33	All admissions	63	46.9	RR, SpO2, SBP, PR, GCS, TP
A2[30]	Raita et al	2019	USA	Retrospective	135,470	2.10	All admissions	46	56.8	TP, PR, SBP, DBP, RR, SpO2
A3[19]	Klug et al.	2020	Israel	Retrospective	799,522	1.55	All admissions	55	51.5	TP, HR, SBP, DBP, SpO2
A4[23]	Kwon et al.	2020	South Korea	Retrospective	23,587	3.98	Infectious patients	63	46.1	SBP, RR, TP, HR
A5[37]	Liu et al.	2011	Singapore	Retrospective	100	40.00	All admissions	65	63	RR, PR, SBP, DBP, SpO2, GCS
A6[35]	Taylor et al.	2015	USA	Retrospective	5,278	4.92	Infectious patients	65	55	HR, TP, SBP, DBP, RR, SpO2
A7[18]	Levin et al.	2017	USA	Retrospective	172,726	0.40	All admissions	46	44.7	TP, HR, RR, SBP, SpO2
A8[20]	Coslovsky et al.	2015	Switzerland	Prospective	7,680	4.60	All admissions	58	62	SBP, HR, SpO2, TP, GCS
A9[34]	Perng et al.	2019	Taiwan	Retrospective	42,220,	4.71	Infectious patients	64	56.5	SBP, GCS, TP, HR
A10[36]	Shamout et al.	2020	UK	Retrospective	37,284	0.80	All admissions	68	48.8	HR, SBP, RR, TP, SpO2

HR: heart rate; PR: pulse rate; RR: respiratory rate; SpO2: oxygen saturation; SBP: systolic blood pressure; DBP: diastolic blood pressure; TP: body temperature: GCS: glasgow coma scale

Table 5. Selected articles list and their ML-related characteristics

Id	ML algorithms	Evaluation Metrics	Time Horizon	Personalized	Handling missing values	Hyperparameter Optimization	Approach	Validation
A1	LR, RF, SVM, NN	AUC, Brier Score	-	No	No	Yes	Binary classification	External
A2	LR, RF, GB DT, DNN	AUC, Sensitivity, Specificity	-	No	No	Yes	Binary classification	Internal
A3	GB	AUC, Sensitivity, Specificity, Brier Score	-	No	No	No	Binary classification	Internal
A4	GB, RF	AUC	72h	No	No	Yes	Binary classification	External
A5	NN, SVM	Accuracy,	9-12	No	No	No	Binary	Internal

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Section and Topic	Item #	Checklist item	Location where item is reported

		Sensitivity, Specificity	minutes				classification	
A6	RF, DT, LR	AUC	-	No	Yes	No	Binary classification	Internal
A7	RF	AUC	-	No	No	No	Binary classification	Internal
A8	LR	AUC, Brier Score	-	No	Yes	No	Binary classification	Internal
A9	KNN, SVM, RF, DNN	AUC, Accuracy	72h, 28 days	No	Yes	No	Binary classification	Internal
A10	DNN	AUC	2h, 24h	Yes	Yes	No	Time series regression	External

ML: machine learning, LR: logistic regression, SVM: support vector machine, KNN: K-Nearest neighbors, DT: decision tree, RF: random forest, GP: gaussian process, NN: neural networks, AUC: area under the curve

Risk of bias in studies	18	**Table 7. Risk of bias (ROB) assessment of included studies according to PROBAST checklist**	Page 16

ID	PROBAST items				
	Participants	Predictors	Outcomes	Sample size & missing data	Statistical analysis
A1	+	+	+	-	-
A2	+	+	+	-	-
A3	+	?	+	-	-
A4	+	?	+	-	-
A5	-	?	+	-	-
A6	+	+	?	+	?
A7	+	+	+	-	-
A8	+	?	?	+	-
A9	+	?	?	-	?
A10	+	+	+	+	?

Results of individual studies	19	There is only one outcome: in-hospital mortality.	Page 14
Results of syntheses	20a	-	
	20b	-	
	20c	-	
	20d	-	

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
Reporting biases	21	-	
Certainty of evidence	22	-	
DISCUSSION			
Discussion	23a	Limitations of included studies This review consisted of ten studies that developed ML-based models to predict mortality in EDs. We found many methodological and reporting shortcomings, and this can lead to poorer performance than reported when the models are deployed in real life. Many studies skipped or failed to report important phases such as data preprocessing, handling missing values, and model development. It has been stated that up to 80% of data analysis is spent on data cleaning and data preparation, so these phases are very crucial for having a ML predictive model with good performance. It also could be one of the reasons why some researchers have discussed poor effectiveness of ML predictive models in clinical environments. Moreover, despite the dramatic advances in ML and introducing novel and effective models, various researchers still have applied traditional ML techniques such as LR and DT. For instance, we found that LR is the second most common algorithm used in the studies. Most ML models have parameters that needed to be tuned to gain the best performance and generalization power and avoiding overfitting, but most of studies did not provide information about the models' validation process and finding the hyperparameters of the models. Of ten studies, nine studies are retrospective studies and only three studies have evaluated their models based on external validation. The ultimate aim of included studies is to predict mortality in EDs ahead of time to provide enough time for clinicians to intervene and prevent mortality. However, only four studies have provided their time horizon, the time that clinicians have to treat the patients. The other studies have built binary classification algorithms to stratify patients in two classes, and their models do not have the capability to be used in practice. Clinical considerations and prospective A recent systematic review on the value of vital sign trends in monitoring and predicting clinical deterioration showed lack of research and knowledge regarding the importance of vital signs in clinical deterioration of ED patients. Even though considering the vital sign as cheap and available clinical predictors, there is a hypothesis stating the lack of evidence to support the usefulness of continuous monitoring of vital signs as a daily routine of clinical practice. The other barrier refers to comparison between ML methods and conventional logistic regression for clinical prediction models. A recent systematic review showed no performance benefits of ML methods over logistic regression. Therefore, further studies need to be performed to prove superiority of ML algorithms over conventional models. It seems that there is still a long way to go for having ML algorithms as the choice of clinical deterioration in EDs. The increase of studies in this field during the last few years, demonstrates an increased attention to appraise the capabilities of ML methods in clinical practice.	Page 7
	23b	It was not possible to conduct meta-analysis.	
	23c	-	
	23d	Perhaps in this research area, the main challenge is integration of ML models into clinical practice. Despite the great research successes in building ML-based predictive models for clinical practice, there remain few examples of ML models being successfully integrated into the daily routine or critical parts of clinical environments. This reveals that what is being done in research is not completely in line with the realities of the clinical practice. In this section, we present recommendations on how to address the most prominent issues and encourage researchers to take them into account when building ML-based predictive models for EDs. Our analysis indicated that most studies are retrospective. However, the impact of ML techniques in clinical environment will need further validation in randomized control trials and prospective studies before widespread clinical adoption where the data could be incomplete and noisy with high level of uncertainty. Recommendation 1: we recommend that instead of building models using historical collected data, researchers consider prospective analysis and think about building adaptive models that can monitor patients in real-time. Recommendation 2: we recommend that researchers investigate the performance of their models using external validation to have a better estimation of the	Page 8

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
		generalization power of their models. Recommendation 3: regarding the preprocessing phase, since some steps such as data cleaning, handling missing and noisy values are an inevitable part of each data analysis study, we recommend that researchers instead of ignoring records with missing values, use proper techniques to handle noisy and incomplete records. For instance, we recommend that complete case analysis is avoided and proper imputation techniques such as iterative multivariate imputation technique is used instead; this is one of the best approaches for imputing missing values. Another challenge in most studies is that the number of event patients is too low (<6%) compared to non-event patients. This can lead to bias and overfitting during developing ML models, so proper under/oversampling techniques such as SMOTE and its variations should be utilized to make a balance between classes before developing models. Another issue is time horizon for prediction. Gerry et al. recommended that time horizon should be limited to a few days at most, since signs of deterioration will probably not be observed for more than a few days. Recommendation 4: we suggest that researchers in this area choose a realistic time horizon rather than prediction of mortality after six or twelve months. Moreover, most of the included studies are based on building a single model for the whole population. Recommendation 5: as patients have different characteristics and norms, we recommend that patients' characteristics, vital signs trajectories, trends, etc. are used for each individual patient to tune ML models to have a better performance in practice. Recommendation 6: we recommend that researchers find a way to make their results and workflow understandable and interpretable by humans. For instance, deep learning is one of the hottest research topics today, and it has been applied in different applications of clinical practice, but it has inherent limitations. For example, despite impressive results, it is not indicative of high-level reasoning and rationality. Such methods are often considered as a "black box", where the decision logic is presented in millions of numerical weights and biases. This lack of transparency could have some legal and ethical implications and might increase distrust of ML models by patients and clinicians, and it might be one of the reasons that patients are likely to trust a clinician more than a machine. This has been described as a crucial problem for ML acceptance in clinical practice and attempts are underway to create more human interpretable models. For example, with the help of image processing techniques, it is possible now to visualize the input and output of ML models; this makes the understanding of ML models easier. Lack of widely accepted and utilized reporting or publication guidelines for implementing ML in clinical practice is challenge that brings difficulty to the research quality assessment, especially for clinicians without a strong mathematics and computer science background. Another challenge for ML models is that they need huge amount of labeled data to be trained, and the performance of ML models often depends on the performance of the human labelling the data. Recommendation 7: as this area is interdisciplinary in nature, it is imperative that clinicians and artificial intelligence experts become better at interdisciplinary collaboration to become more familiar with the needs, limitations, and challenges in clinical and technical domains, which ultimately increases the quality of the predictive models. Recommendation 8: we recommend that researchers should be careful and accurate when reporting their studies. It has been shown that the quality of reporting of predictive models in this clinical domain is poor. Only with full and transparent reporting of information about all aspects of a predictive model, the risk of bias and the usefulness of that model can be assessed. It is recommended that researchers use available checklists such as SPIRIT-AI for designing the trial, and TRIPOD and CONSORT-AI for reporting the findings to make sure that they provide sufficient information to clearly and accurately report their studies.	
OTHER INFORMATION			
Registration and protocol	24a	-	
	24b	-	
	24c	-	
Support	25	There is no funding attached to this systematic review.	Page 11
Competing interests	26	None declared.	Page 11
Availability of data, code and other materials	27	All data relevant to the study are included in the article or uploaded as online supplementary information.	Page 11

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From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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BMJ Open

Machine Learning Techniques for Mortality Prediction in Emergency Departments: A Systematic Review

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-052663.R1
Article Type:	Original research
Date Submitted by the Author:	28-Aug-2021
Complete List of Authors:	Naemi, Amin; University of Southern Denmark, The Maersk Mc-Kinney Moller Institute Schmidt, Thomas; University of Southern Denmark, The Maersk Mc-Kinney Moller Institute Mansourvar, Marjan; University of Southern Denmark, The Maersk Mc-Kinney Moller Institute Naghavi-Behzad, Mohammad ; University of Southern Denmark, Department of Clinical Research; Odense University Hospital, Department of Nuclear Medicine Ebrahimi, Ali; University of Southern Denmark, The Maersk Mc-Kinney Moller Institute Wiil, Uffe ; University of Southern Denmark, Maersk Mc-Kinney Moller Institute
Primary Subject Heading:	Emergency medicine
Secondary Subject Heading:	Health informatics
Keywords:	Health informatics < BIOTECHNOLOGY & BIOINFORMATICS, Information technology < BIOTECHNOLOGY & BIOINFORMATICS, Information management < BIOTECHNOLOGY & BIOINFORMATICS

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Machine Learning Techniques for Mortality Prediction in Emergency Departments: A Systematic Review

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Abstract

Objectives: This systematic review aimed to assess the performance and clinical feasibility of ML algorithms in prediction of in-hospital mortality for medical patients using vital signs at emergency departments.

Design: A systematic review was performed.

Setting: The databases including Medline (PubMed), Scopus, and Embase (Ovid) were searched between 2010 and 2021, to extract published articles in English, describing ML-based models utilizing vital signs variables to predict in-hospital mortality for patients admitted at emergency departments. CHARMS checklist was used for study planning and data extraction. The risk of bias for included papers was assessed using the PROBAST tool.

Participants: Admitted patients to the ED

Main outcome measure: In-hospital mortality.

Results: Fifteen articles were included in the final review. We found that eight models including logistic regression, decision tree, K-nearest neighbors, support vector machine, gradient boosting, random forest, artificial neural networks, and deep neural networks have been applied in this domain. Most studies failed to report essential main analysis steps such as data preprocessing and handling missing values. Fourteen included studies had a high risk of bias in the statistical analysis part, which could lead to poor performance in practice. Although the main aim of all studies was developing a predictive model for mortality, nine articles did not provide a time horizon for the prediction.

Conclusion: This review provided an updated overview of the state-of-the-art and revealed research gaps; based on these, we provide eight recommendations for future studies to make the use of ML more feasible in practice. By following these recommendations, we expect to see more robust ML models applied in the future to help clinicians identify patient deterioration earlier.

Keywords: Machine learning, emergency department, mortality, prediction, healthcare, vital signs.

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Strength and limitations of this study

- To the best of our knowledge, this is the most recent systematic review on the topic of using machine learning algorithms to predict mortality at emergency departments.
- All included studies were evaluated regarding potential bias and the analysis was done following standard checklists.
- This review highlights the research gaps and important barriers that avoid the integration of ML into clinical practice and provides recommendations to address these challenges in future research.
- It was not possible to conduct meta-analysis, due to the heterogeneity of included studies.
- Only studies with vital signs among their predictors were included in the review.

1 INTRODUCTION

The improved life expectancy in Europe and most developed countries[1] has increased admissions to the Emergency Departments (ED)s, especially for elderly people who have a higher risk of morbidity and mortality.[2,3] Also, having more inter- and intra-city travels and organized emergency medical services to transfer these patients to ED brought more importance as well as patient load to EDs.[4,5] The average rate of mortality among in-patient admissions is around 2% in United States,[6] indicating the necessity of using clinical deterioration prediction models.[7] Moreover, a recent study has shown that some in-hospital deaths due to clinical deterioration can be avoided.[8]

Vital signs including heart rate (HR), pulse rate (PR), respiratory rate (RR), oxygen saturation (SpO2), blood pressure (BP), body temperature (TP), and Glasgow coma scale (GCS) are essential parts of the patients' monitoring process. The data received from these parameters could be considered as the cheapest and the most important information, which could easily be collected during admission.[9] Although vital signs are among the first data that clinicians observed to have an initial assessment about the patients' condition and have been used in clinical practice for over a century, few studies have effectively quantified their performance in various clinical applications.[10] In recent years, various studies have emphasized that changes in vital signs happen several hours before a serious adverse event.[11–13] Nowadays, vital signs play a key role in identification of patients with risk of deterioration in ED, however deterioration often occurs unnoticed or is not identified until it becomes too late to intervene.[9]

Machine Learning (ML) addresses the question of how to build a computer model that automatically enhances performance through experiments, lying at the intersection of computer science, statistics, and a variety of other disciplines concerned about automatic improvement over time, and inferring and decision making under uncertainty.[14] Various ML algorithms have been developed to solve a wide variety of problems. The most commonly used ML

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algorithms are supervised learning algorithms focusing on relationships and dependencies between the input and output features.[15] Hence, we can identify the output values for unseen data based on the relationships learned from the training datasets. Unsupervised learning is subset of ML that is used in pattern recognition and descriptive modeling without the use of output categorizes or data labeling.[16]

Because of enormous advancements in the development of modern ML algorithms such as deep learning, the availability of large databases, and increased computational power, ML has progressed rapidly over the last two decades from laboratory curiosity to practical technology in a wide range of commercial applications. Consequently, utilization of ML techniques in the healthcare domain is increasing rapidly, and the list of tasks where ML has matched or outperformed physicians is growing. Clinical outcome prediction is one challenge receiving attention, and by improvement of ML techniques, the ML-based systems aim to outperform conventional clinical scoring systems.[14,17]

Various studies have been made in this area evaluating the prediction accuracy of in-hospital mortality, cardiac arrest, and intensive care unit (ICU) transfer. To reach this goal, various techniques such as logistic regression (LR), support vector machine (SVM), K-nearest neighbors (KNN), decision tree (DT), random forest (RF), Gaussian process (GP), and artificial neural networks (NN) have been applied, and the results confirmed the appropriateness of using ML in this area.[14] The capability of handling large datasets is the most considerable advantage of ML algorithms, which can result in more accurate prediction of clinical outcomes. Currently, we have various prediction models, using different techniques, in order to improve the triage process and to provide better identification of high-risk patients.[18,19] Although several studies based on different models were conducted, no single model has yet proved to be superior regarding clinical efficacy.[20–24]

Despite the use of vital signs in patient monitoring and clinical decision making, the importance of specific types of vital signs, their correlation, and frequency of registration to best prevent adverse events and in-hospital mortality is still unclear.[9] However, ML could be a good solution for these challenges. Considering the possible role of ML models in prediction of clinical deterioration of patients in EDs,[25] it is essential to analyze and merge the results of previous studies to have an updated overview of the state-of-the-art. This systematic review focused on the clinical efficacy and technical implications of ML-based models in prediction of mortality using vital signs predictors in patients admitted to EDs.

2 MATERIALS AND METHODS

This systematic review was conducted in the period from July 2020 to August 2021. The main questions addressed by this study are presented in Table 1.

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Table 1. Research questions	
Q1	What ML techniques have been used to predict in-hospital mortality in EDs?
Q2	What are the common vital signs variables used in studies?
Q3	How researchers prepared data for ML techniques?
Q4	What are the approaches to solve the problem (e.g. binary classification or time series prediction)?
Q5	What are the challenges and open issues in this domain?

2.1 Search strategy and study selection

Three databases including Medline (PubMed), Scopus, and Embase (Ovid) were targeted. Relevant articles were extracted using a broad range of relevant keywords. We stratified keywords into five groups namely, ML keywords, medical keywords, document type, publication year, and language. The keywords in a group were paired using OR operators and all groups were paired using AND operator. Table 2 shows the search keywords that were applied to the titles, abstracts, and full text in the three databases.

Table 2. Search keywords in different groups	
Group 1 – ML keywords	Artificial intelligence, machine learning, deep learning, learning algorithms, supervised machine learning, unsupervised machine learning
Group 2 – Medical keywords	Clinical deterioration, mortality, in-hospital mortality, death, vital sign, emergency departments
Group 3 – Document type	Journal
Group 4 – Publication year	1st January 2010 to 1st October 2020
Group 5 – Language	English
Final result	(Group 1) AND (Group 2) AND (Group 3) AND (Group 4) AND (Group 5)

Inclusion and exclusion criteria are listed in Table 3. Our queries for three databases can be found in supplementary material as S1 file.

Table 3. Inclusion and exclusion criteria	
Inclusion Criteria	**Exclusion criteria**
Prediction of in-hospital mortality prediction should be the main aim of the study.	Studies which are conference articles, posters, abstracts, books or book chapters, and review articles.
The study should be done at ED.	Studies which were done in pediatrics field.
The study should be done on adult patients.	Non-English studies
ML algorithms should be used for the prediction task.	
Vital signs variables should be among the predictors used to build ML models.	

2.2 Data extraction

Three researchers (AN, MM, AE) screened the title and extracted abstracts, independently. The screening process was done using the Covidence tool. Then, two researchers (AN, TS) read the full texts, independently resolving the disagreement by supervision of the senior researcher (UKW). Spreadsheets for item extraction were prepared based on the critical appraisal and data extraction for systematic reviews of prediction modelling studies (CHARMS) checklist.[26] The extracted items include study design, publication date, source of data, study population, outcomes, time horizon, considered vital signs predictors, data preprocessing, model development, model performance, model validation and evaluation, and the interpretation.

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2.3 *Risk of bias assessment*

Risk of bias (ROB) for each study was assessed by using the prediction risk of bias assessment tool (PROBAST) checklist[27] and reported based on an adapted form.[28] Extracted articles were classified into three different categories (low, high, unclear). A study is considered as having a high ROB if the study has at least one high ROB in four domains. ROB assessment of each study was checked by two of our researchers (AN, MNB); disagreements were resolved in collaboration with a senior researcher (UKW).

2.4 *Patient and public involvement*

No patient involved.

3 RESULTS

The initial search resulted in 7,466 records, which after removing duplicates, evaluation of eligibility for inclusion, full-text assessment, quality assessment and review of the references of the included papers, was narrowed down to fifteen studies. The study selection flowchart is shown in Figure 1.

Figure 1. Flow diagram of study selection (PRISMA chart).

General information, as well as study design and population of included studies, are summarized in Table 4. Twelve out of fifteen articles were published after 2019, which indicates an increased interest recently among researchers to explore the capability of ML in prediction of clinical outcomes in EDs.

3.1 *Studies design*

Among extracted articles, fourteen studies have used electronic data of the related hospitals, while one of the studies was based on registry-based public data.[29] One of the essential parts in utilizing ML techniques in each application is gathering and preparation of the dataset, which has a high impact on the performance of the final ML-based system. In general, ML techniques need an extensive dataset to produce good results, and the performance of ML models depends largely on the number of training examples.[14] Based on Riley et al.,[30] thirteen studies had enough number of patients to build acceptable models through their datasets, while two studies proposed a model based on only 100 and 445 patients.[31,32]

3.2 *Vital signs predictors*

We considered studies, which utilized vital signs among their predictors in building models. The vital signs set used in extracted articles included HR, PR, RR, SpO2, BP, TP, and GCS (Table 4). BP and TP were the most widely used vital sign predictors among the included studies. SpO2, RR, and HR were common vital signs, which were used in thirteen,

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thirteen, and eleven articles, respectively, while GCS and PR were the least commonly used vital signs. Besides the vital signs, all studies used other clinical predictors such as demographic variables (e.g. age, sex, race, marital status),[18,19,21,22,29,31–40] arrival mode (walk-in and ambulance),[18,19,29,32,36] blood tests (e.g. albumin, creatinine, hemoglobin, potassium, sodium, white cell count, urea),[21,33,35–37,39] electrocardiogram signal,[31] chief complaints,[18,19,29,34,36,38,40] length of stay at hospital,[22] medications,[19,32,36–38] comorbidities,[36] diagnoses,[22,38] medical history,[18,36–38] and triage information.[19,21,36,38]

Table 4. General characteristics of included studies

ID	Authors	Year	Country	Study type	Population	Outcome portion (%)	Patients type	Mean age (y)	Male (%)	Vital signs
A1[18]	Levin et al.	2017	USA	Retrospective	172,726	0.40	All admissions	46	44.7	TP, HR, RR, SBP, SpO2
A2[19]	Klug et al.	2020	Israel	Retrospective	799,522	1.55	All admissions	55	51.5	TP, HR, SBP, DBP, SpO2
A3[21]	Faisal et al.	2020	UK	Retrospective	24,696	5.33	All admissions	63	46.9	RR, SpO2, SBP, PR, GCS, TP
A4[22]	Kwon et al.	2020	South Korea	Retrospective	23,587	3.98	Infectious patients	63	46.1	SBP, RR, TP, HR
A5[29]	Raita et al	2019	USA	Retrospective	135,470	2.10	All admissions	46	56.8	TP, PR, SBP, DBP, RR, SpO2
A6[31]	Liu et al.	2011	Singapore	Retrospective	100	40.00	All admissions	65	63	RR, PR, SBP, DBP, SpO2, GCS
A7[32]	Karlsson et al.	2021	Sweden	Retrospective	445	14.1	Infectious patients	73	52.6	HR, SBP, DBP, RR, TP, SpO2
A8[33]	Chen et al.	2021	Taiwan	Retrospective	52,626	9.4	Infectious patients	72	58.3	HR, SBP, DBP, TP, RR, SpO2, GCS
A9[34]	Joseph et al.	2020	USA	Retrospective	445,925	< 13	All admissions	53	45.9	HR, SBP, DBP, RR, TP, SpO2
A10[35]	Rodriguez et al.	2021	Columbia	Retrospective/ Prospective	2,510	11.5	Infectious patients	62	49.8	SBP, DBP, TP, RR, SpO2, GCS
A11[36]	Soffer et al.	2020	Israel	Retrospective	118,262	5.3	All admissions	73	52.6	HR, SBP, DBP, TP, RR, SpO2
A12[37]	van Doorn et al.	2021	Netherlands	Retrospective	1,344	13	Infectious patients	71	54.4	HR, SBP, DBP, TP, RR, SpO2, GCS
A13[38]	Taylor et al.	2015	USA	Retrospective	5,278	4.92	Infectious patients	65	55	HR, TP, SBP, DBP, RR, SpO2
A14[39]	Perng et al.	2019	Taiwan	Retrospective	42,220,	4.71	Infectious patients	64	56.5	SBP, GCS, TP, HR
A15[40]	Shamout et al.	2020	UK	Retrospective	37,284	0.80	All admissions	68	48.8	HR, SBP, RR, TP, SpO2

HR: heart rate; PR: pulse rate; RR: respiratory rate; SpO2: oxygen saturation; SBP: systolic blood pressure; DBP: diastolic blood pressure; TP: body temperature: GCS: Glasgow coma scale

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3.3 Data preparation

Real-world datasets often need to be transformed, cleaned or changed before use. These datasets may contain missing values, noise, and even wrong entries or inconsistencies.[41] Missing values issue is one of the main challenges in clinical data mining[42] that often provide additional information about patients. Therefore, in almost all cases, raw data will need to be preprocessed before it can be used as the basis for ML modeling. This preprocessing may affect the final model, and the specific steps are important to clarify. However, ten articles did not utilize proper techniques to impute missing values and excluded up to 32% of incomplete patients' records in some studies.[21,29] Different approaches, including replacing missing values with median,[33,39] considering missing values as a special value,[38] and imputation using Gaussian process technique[40] were applied to impute missing values.

3.4 Outcome and time horizon

The percentage of target patients in the nine included studies is less than 6%. It should be noted that Liu et al.[31] and Karlsson et al.[32] have conducted their studies on small datasets containing 100 and 445 patients from which 40 and 63 patients died during hospitalization, respectively. Since the outcome patients in most studies were few, large datasets should be collected to obtain enough event patients.

As the main aim of the included studies was building ML models to predict mortality at EDs, these articles should specify how much earlier their models can predict mortality. However, as shown in Table 5, only six studies reported their time horizon for mortality prediction. 9-12 minutes, 2 hours, 24 hours, 72 hours, 7 days, 28 days, 30 days, and 31 days ahead were reported as time horizons. Since studies in this area try to predict mortality in advance, all studies must report their time horizons, meaning the time that their models provide clinicians to intervene before mortality occurs.

3.5 Imbalanced data

Fourteen studies utilized classification approaches to predict mortality at EDs. Since the number of patients who died during hospitalization (Table 4) for many studies is small, some proper preprocessing techniques such as balancing classes are needed. In addition, except,[40] all other studies stratified patients into two groups, e.g. non survived patients and survived patients, but the number of patients who died during hospitalization was much lower. This challenge which is called imbalanced data, can affect the performance of ML models significantly; thus, proper techniques should be applied to address this challenge. Nevertheless, only two studies[22,33] mentioned this challenge and used Synthetic Minority Over-sampling Technique (SMOTE)[43] to address it.

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Table 5. Selected articles list and their ML-related characteristics

Id	ML algorithms	Evaluation Metrics	Time Horizon	Personalized	Handling missing values	Hyperparameter Optimization	Approach	Validation
A1	RF	AUC	-	No	No	No	Binary classification	Internal
A2	GB	AUC, Sensitivity, Specificity, Brier Score	-	No	No	No	Binary classification	Internal
A3	LR, RF, SVM, NN	AUC, Brier Score	-	No	No	Yes	Binary classification	External
A4	GB, RF	AUC	72h	No	No	Yes	Binary classification	External
A5	LR, RF, GB DT, DNN	AUC, Sensitivity, Specificity	-	No	No	Yes	Binary classification	Internal
A6	NN, SVM	Accuracy, Sensitivity, Specificity	9-12 minutes	No	No	No	Binary classification	Internal
A7	RF	AUC, Sensitivity, Specificity, PPV, NPV, PLR, NLR	7days, 30 days	No	No	No	Binary classification	Internal
A8	SVM, GB, NN	AUC, accuracy, Sensitivity, Specificity, PPV, NPV	-	No	Yes	No	Binary classification	Internal
A9	LR, DNN, GB	AUC, accuracy, Sensitivity, Specificity,	-	No	No	Yes	Binary classification	Internal
A10	DT, RF, NN, SVM	AUC, accuracy	-	No	No	No	Binary classification	Internal
A11	GB	AUC, Sensitivity, Specificity, NPV, PPV, FPR	-	No	Yes	No	Binary classification	Internal
A12	LR, NN, RF, GB	AUC, Sensitivity, Specificity	31 days	No	No	No	Binary classification	Internal
A13	RF, DT, LR	AUC	-	No	Yes	No	Binary classification	Internal
A14	KNN, SVM, RF, DNN	AUC, Accuracy	72h, 28 days	No	Yes	No	Binary classification	Internal
A15	DNN	AUC	2h, 24h	Yes	Yes	No	Time series regression	External

ML: machine learning, LR: logistic regression, SVM: support vector machine, KNN: K-Nearest neighbors, DT: decision tree, RF: random forest, GP: gaussian process, NN: neural networks, DNN: deep neural networks, AUC: area under the curve, PPV: positive predictive value, NPV: negative predictive value, PLR: positive likelihood ratio, NLR: negative likelihood ratio, FPR: false positive rate.

3.6 Machine Learning models

As shown in Table 5, eight ML techniques have been utilized in the included studies. RF was the most common ML technique and was used in nine studies. A brief description and list of advantages and disadvantages of each ML technique is summarized in Table 6. Although most studies have been made during the last five years, we observed traditional algorithms such as LR more than advanced algorithms, applied to build predictive models in five studies.

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Table 6. Summary of Machine Learning algorithms' description, advantages and disadvantages

Algorithm	Description	Advantages	Disadvantages
LR[44]	LR is a supervised ML algorithm adopted from linear regression. It can be used for classification problems and finding the probability of an event happening.	Fast training, good for small datasets, easy to understand.	Not very accurate, not proper for nonlinear problems, high chance of overfitting, not flexible to adopt to complex datasets.
DT[45]	DT is a supervised ML algorithm that solves a problem by transforming the data into a tree representation where each internal node represents an attribute, and each leaf denotes a class label.	Easy to understand and interpret, robust to outliers, no standardization or normalization required, useful for regression and classification.	High chance of overfitting, not suitable for large datasets, adding new samples lead to regeneration of the whole tree.
KNN[46]	KNN is a supervised and instance-based ML algorithm. It can be used when we want to forecast a label of a new sample based on similar samples with known labels. Different similarity or distance measures such as Euclidean can be used.	Simple and easy to understand, easy to implement, no need for training, useful for regression and classification.	Memory intensive, costly, slow performance, all training data might be involved in decision making.
SVM[47]	SVM is an instance-based and supervised ML technique that generates a boundary between classes known as hyperplane. Maximizing the margin between classes is the main goal of this technique.	Efficient in high dimensional spaces. Effective when the number of dimensions is greater than the number of samples, long training time, useful for regression and classification.	Not suitable for large datasets, not suitable for noisy datasets, regularization capabilities which prevent overfitting, handling non-linear data.
GB[48]	GB is a supervised ML algorithm, which produces a model in the form of an ensemble of weak prediction models, usually DT. GB is an iterative gradient technique that minimizes a loss function by iteratively selecting a function that points towards the negative gradient.	High accuracy, high flexibility, fast execution, useful for regression and classification, robust to missing values and overfitting.	Sensitive to outliers, not suitable for small datasets, many parameters to optimize.
RF[49]	RF is an ensemble and supervised ML algorithm that is based on the bagging technique, which means that many subsets of data are randomly selected with replacement and each model such as DT is trained using one subset. The output is the average of all predictions of various single models.	High accuracy, fast execution, useful for regression and classification, robust to missing values and overfitting.	Not suitable for limited datasets, may change considerably by a small change in the data.
NN[50]	NN is a family of supervised ML algorithms. It is inspired by biological neural network of the human brain. NN consists of input, hidden, output layers and multiple neurons (nodes) carry data from input layer to output layer.	Accurate, suitable for complex, non-linear classification, and regression problems.	Very slow to train and test, requires large amount of data, computationally expensive and prone to overfitting.
DNN[51]	DNN is a family of supervised ML algorithms. DNN is based on NN where the adjective "deep" comes from the use of multiple layers in the network. Usually having two or more hidden layers counts as DNN. There are some specific training algorithms and architecture such as LSTM, GAN, CNN for DNNs. DNNs provide the opportunity to solve complex problems when the data is very diverse, unstructured, and inter-connected.	High accuracy, features are automatically deduced and optimally tuned, robust to noise, architecture is flexible.	Needs very large amount of data, computationally expensive, not easy to understand, no standard theory in selecting the right settings, difficult for less skilled researchers.

LR: Logistic Regression, DT: Decision Tree, KNN: K-Nearest Neighbors, SVM: Support Vector Machine, GB: Gradient Boosting, RF: Random Forest, NN: Neural Networks, LSTM: long-short term memory networks, GAN: generative adversarial networks, CNN: convolutional neural networks, DNN: Deep Neural Networks.

3.7 Validation and evaluation

Ten articles used k-fold cross-validation,[21,22,29,31,32,34,35,37–39] while one study applied bootstrapping technique for validation purpose.[36] Two studies split their data into training and test sets and used a portion of the training set for validation[19,40]. Three studies investigated the generalization capability of their models using external validation, while the rest of the studies were internally validated. Researchers have used different metrics for evaluation of ML models in predicting mortality at EDs. The area under the curve (AUC) metric was used in fourteen articles. Eight studies assessed

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the performance of ML models using sensitivity and specificity. Five studies presented the accuracy of models, while positive predictive value (PPV) and negative predictive value (NPV) were used in three studies. The Brier Score, positive likelihood ratio (PLR), negative likelihood ratio (NLR), and false positive rate (FPR) were less common metrics used in two studies, one, one, and one study, respectively.

3.8 Personalized model

Patients' characteristics should be considered in developing predictive models and the potential impact on the improvement of models' performance should be investigated. However, among the included studies, only one study used information of each patient to predict the risk of mortality for the same patient.[40] They considered patients' vital sign trajectories as time series data and introduced a ML model to follow up the trends of vital signs for each patient and by that they tried to consider patients' characteristics, while the other studies only build one model for the whole population and categorized patients into two groups and binary classifiers using different ML algorithms were built.

3.9 Risk of bias assessment

There is a significant amount of incomplete reporting within the studies' results. For instance, most studies did not report information about the tuning process of ML models and hyperparameters values considered during the model training process. A summary of the PROBAST assessment is shown in Table 7. As shown, participants' selection was done properly except in two studies. However, most studies had a high ROB in the statistical analysis domain. Among the studies, only Joseph et al.[34] utilized transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD)[52] guideline and provide adequate information about feeding data to models and architecture of implemented NN, so only this study was considered as low ROB in the statistical domain.

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Table 7. Risk of bias (ROB) assessment of included studies according to PROBAST checklist

ID	PROBAST items				
	Participants	Predictors	Outcomes	Sample size & missing data	Statistical analysis
A1	+	+	+	-	-
A2	+	?	+	-	-
A3	+	+	+	-	-
A4	+	?	+	-	-
A5	+	+	+	-	-
A6	-	?	+	-	-
A7	-	+	+	-	-
A8	+	+	+	+	-
A9	+	+	+	?	+
A10	?	+	+	-	?
A11	+	+	+	?	?
A12	?	+	+	-	?
A13	+	+	?	+	?
A14	+	?	?	-	?
A15	+	+	+	+	?

+: low risk of bias, -: high risk of bias, ?: unclear risk of bias

4 DISCUSSION

4.1 Weaknesses and strengths of included studies

This review consisted of fifteen studies that developed ML-based models to predict mortality in EDs. Our analysis showed that most studies described the datasets in sufficient detail and had enough number of participants. Moreover, the outcome domain had the lowest risk of bias which shows most studies had a clear outcome definition. However, we found several methodological and reporting shortcomings, and this can lead to poorer performance than reported when the models are deployed in real life. Many studies skipped or failed to report important phases such as data preprocessing, handling missing values, and model development. It has been stated that up to 80% of data analysis is spent on data cleaning and data preparation,[41] so these phases are very crucial for having a ML predictive model with good performance. It also could be one of the reasons why some researchers have discussed the poor effectiveness of ML predictive models in clinical environments.[53,54] Moreover, despite the dramatic advances in ML and introducing novel and effective models, various researchers still have applied traditional ML techniques such as LR and DT. For instance, we found that LR is the third most common algorithm used in the studies. Most ML models have parameters that needed to be tuned to gain the best performance and generalization power and avoiding overfitting, but most studies did not provide information about the models' validation process and finding the hyperparameters of the models. Of

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fifteen studies, fourteen studies are retrospective studies and only three studies have evaluated their models based on external validation.

The ultimate aim of included studies was to predict mortality in EDs ahead of time to provide enough time for clinicians to intervene and prevent mortality. However, only six studies have provided their time horizon, the time that clinicians have to treat the patients. The other studies have built binary classification algorithms to stratify patients into two classes, and their models do not have the capability to be used in practice.

4.2 Clinical considerations and prospective

A recent systematic review on the value of vital sign trends in monitoring and predicting clinical deterioration showed a lack of research and knowledge regarding the importance of vital signs in clinical deterioration of ED patients.[9] Even though considering the vital sign as cheap and available clinical predictors, there is a hypothesis stating the lack of evidence to support the usefulness of continuous monitoring of vital signs as a daily routine of clinical practice.[55–57] The other barrier refers to the comparison between ML methods and conventional LR for clinical prediction models. A recent systematic review showed no performance benefits of ML methods over logistic regression.[58] However, as mentioned in the previous section, this conclusion could be due to lack of enough attention to the important stages and practical issues such as data preparation or models' tuning, and consequently results are not satisfactory and implemented models are not robust. Therefore, further studies need to be performed to prove the superiority of ML algorithms over conventional models. It seems that there is still a long way to go for having ML algorithms as the choice of clinical deterioration in EDs. The increase of studies in this field during the last few years proves an increased attention to appraise the capabilities of ML methods in clinical practice.

4.3 Challenges and recommendations for future work

Perhaps in this research area, the main challenge is integration of ML models into clinical practice. A recent systematic review based on performance metrics, particularly AUC, showed that ML predictive models outperformed usual care in most detection and prediction tasks at ED. However, the authors mentioned that many studies have limited applicability to clinical practice and there are other considerations more than performance metrics as well as barriers that should be taken into account to have successful ML models in real life.[59] Therefore, despite the great research successes in building ML-based predictive models for clinical practice, there remain few examples of ML models being successfully integrated into the daily routine or critical parts of clinical environments.[60] This reveals that what is being done in research is not completely in line with the realities of clinical practice. In this section, we mention some of these issues

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and technical barriers, then present recommendations on how to address the most prominent issues and encourage researchers to take them into account.

Our analysis indicated that most studies are retrospective. However, the impact of ML techniques in clinical environment will need further validation in randomized control trials and prospective studies before widespread clinical adoption where the data could be incomplete and noisy with high level of uncertainty. *Recommendation 1*: we recommend that besides building models using historical collected data, researchers consider prospective analysis and think about building adaptive models that can monitor patients in real-time.

Although external validation is one of the most rigorous ways to assess the generalization power of predictive models, it is still one of the common approaches for showing the robustness of predictive models.[27,59] *Recommendation 2*: we recommend that researchers investigate the performance of their models using external validation to have a better estimation of the generalization power of their models. *Recommendation 3*: regarding the preprocessing phase, since some steps such as data cleaning, handling missing and noisy values are an inevitable part of each data analysis study, we recommend that researchers instead of ignoring records with missing values, use proper techniques to handle noisy and incomplete records. For instance, we recommend that complete case analysis is avoided and proper imputation techniques such as iterative multivariate imputation technique[61] which is one of the best approaches for imputing missing values is used instead.[62] Another challenge in most studies is that the number of event patients is too low compared to non-event patients. This can lead to bias and overfitting during developing ML models, so proper under/oversampling techniques such as SMOTE and its variations should be utilized to make a balance between classes before developing models.

Another issue is time horizon for prediction. Gerry et al. recommended that time horizon should be limited to a few days at most, since signs of deterioration will probably not be observed for more than a few days.[63] *Recommendation 4*: we suggest that researchers in this area choose a realistic time horizon rather than prediction of mortality after six or twelve months. Moreover, most of the included studies are based on building a single model for the whole population. *Recommendation 5*: as patients have different characteristics and norms, we recommend that patients' characteristics, vital signs trajectories, trends, etc. are used for each individual patient to tune ML models to have a better performance in practice. *Recommendation 6*: we recommend that researchers find a way to make their results and workflow understandable and interpretable by humans. For instance, deep learning is one of the hottest research topics today, and it has been applied in different applications of clinical practice, but it has inherent limitations. For example, despite

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impressive results, it is not indicative of high-level reasoning and rationality. Such methods are often considered as a "black box", where the decision logic is presented in millions of numerical weights and biases. This lack of transparency could have some legal and ethical implications and might increase distrust of ML models by patients and clinicians, and it might be one of the reasons that patients are likely to trust a clinician more than a machine. This has been described as a crucial problem for ML acceptance in clinical practice and attempts are underway to create more human interpretable models. For example, with the help of image processing techniques, it is possible now to visualize the input and output of ML models; this makes the understanding of ML models easier. [17]

Lack of widely accepted and utilized reporting or publication guidelines for implementing ML in clinical practice is a challenge that brings difficulty to the research quality assessment, especially for clinicians without a strong mathematics and computer science background.[17] Another challenge for ML models is that they need a huge amount of labeled data to be trained, and the performance of ML models often depends on the performance of the human labeling the data. *Recommendation 7*: as this area is interdisciplinary in nature, it is imperative that clinicians and artificial intelligence experts become better at interdisciplinary collaboration to become more familiar with the needs, limitations, and challenges in clinical and technical domains, which ultimately increases the quality of the predictive models. *Recommendation 8*: we recommend that researchers should be careful and accurate when reporting their studies. It has been shown that the quality of reporting of predictive models in this clinical domain is poor. Only with full and transparent reporting of information about all aspects of a predictive model, the risk of bias and the usefulness of that model can be assessed. It is recommended that researchers use available checklists such as SPIRIT-AI[64] for designing the trial, and TRIPOD[52] and CONSORT-AI[65] for reporting the findings to make sure that they provide sufficient information to clearly and accurately report their studies.

4.4 Strengths and limitations of this study

To the best of our knowledge, this study is the most recent systematic review on the topic of using ML algorithms to predict mortality at emergency departments. This study was done with the help of an experienced artificial intelligence expert, an experienced research librarian who is familiar with the health research area, and three expert clinicians at ED. Moreover, ROB analysis was done using PROBAST checklist which has four domains that assess different aspects of studies. Data extraction was done using a standard checklist called CHARMS. However, our study has some limitations. First, it was not possible to quantify the analysis or a conduct meta-analysis due to the high heterogeneity of

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included studies. Also, we have considered the vital signs due to high availability at ED, while the role of other clinical predictors in patients' clinical conditions should be investigated.

5 CONCLUSION

The application of ML methods to identify clinical deterioration remains equally challenging as identification of deterioration using track and trigger protocols and similar human-driven protocols. However, since the ML approaches are as diverse as the problem they deal with, assessment of various methods and their performance are needed. This systematic review was performed on the topic of utilization of ML techniques to predict mortality at EDs using vital signs. Our systematic review of the literature provides an updated overview of the state-of-the-art on this topic (covering 2010 to 2021). Initially, 7,466 potential articles were identified of which fifteen were included in the analysis. After a comprehensive review of these fifteen articles, we generated eight recommendations to increase the feasibility of implementing ML models in EDs. These recommendations provide actionable suggestions to be used to increase the quality of future work in this area.

Acknowledgement

The authors would like to thank Dr. Mette Brandt Eriksen, research librarian at University of Southern Denmark library for help with the search strategy and compiling the literature databases, and Prof. Annmarie Touborg Lassen, Prof. John Kellet, and Prof. Mikkel Brabrand, expert emergency department clinicians at Odense University Hospital, for their help during performing this systematic review.

Author contributions: All authors were involved in conception and design of the study. Literature search was conducted by AN under supervision of an expert librarian. AN, TS, MM, and AE participated in screening and extracting relevant studies. AN, MNB, TS, and UKW were responsible for analyzing, and drafting the manuscript. All authors made critical revisions.

Funding: There is no funding attached to this study.

Competing interests: None declared.

Patient consent for publication: Not required.

Data availability statement: Extra data can be accessed via the Dryad data repository at http://datadryad.org/ with the doi.org/10.5061/dryad.8w9ghx3nc.

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Ethics statement: Formal ethical approval is not required, as primary data will not be collected. The review will be disseminated in peer-reviewed publications.

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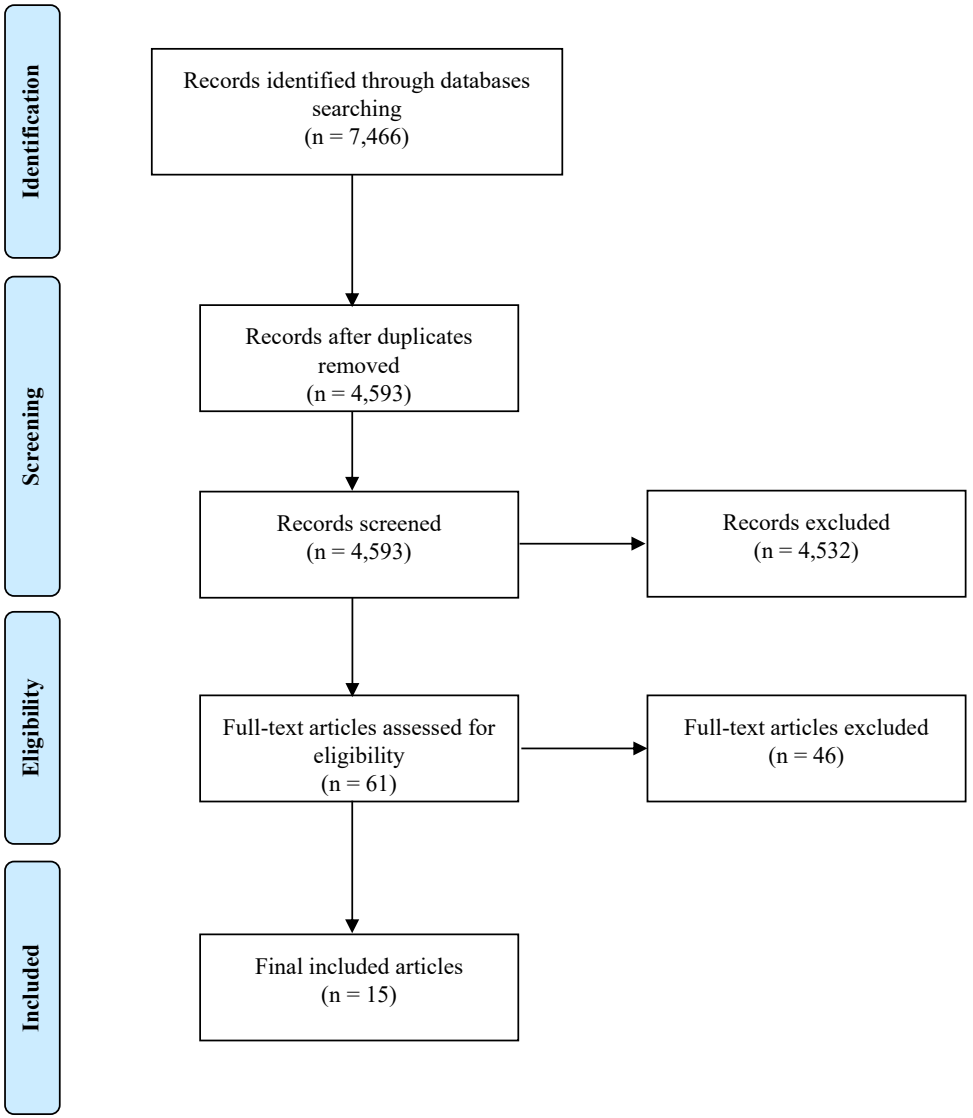
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Identification
Screening
Eligibility
Included
Records identified through databases searching (n = 7,466)
Records after duplicates removed (n = 4,593)
Records screened (n = 4,593)
Records excluded (n = 4,532)
Full-text articles assessed for eligibility (n = 61)
Full-text articles excluded (n = 46)
Final included articles (n = 15)

Scopus: (TITLE-ABS-KEY ((clinical W/2 deteriorat*) OR "mortalit*" or "death" or "hospital mortality" or "vital sign*" or "emergency department") AND TITLE-ABS-KEY ("artificial intelligenc*" or "machine learn*" or "artificial neural network*" or "bayesian learn*" or "classification algorithm*" or "deep learn*" or "fuzzy system*" or "hidden markov model*" or "learning algorithm*" or "supervised machine learn*" or "support vector machine*" or "unsupervised machine learn*"))

Embase: ("Deterioration/" or "hospital mortality/" or "vital signs/" or ((clinical adj3 deteriorat*) or "hospital mortality" or "vital sign*").mp. [mp=ti, ab, hw, tn, ot, dm, mf, dv, kw, fx, dq, nm, kf, ox, px, rx, ui, sy]) and ("artificial intelligence/" or "machine learning" or "artificial neural network/" or "Bayesian learning/" or "classification algorithm/" or "deep learning/" or "fuzzy system/" or "hidden markov model/" or "learning algorithm/" or "supervised machine learning/" or "support vector machine/" or "unsupervised machine learning/" or ("artificial intelligenc*" or "machine learn*" or "artificial neural network*" or "Bayesian learn*" or "classification algorithm*" or "deep learn*" or "fuzzy system*" or "hidden makov model*" or "learning algorithm*" or "supervised machine learn*" or "support vector machin*" or "unsupervised machine learn*").mp. [mp=ti, ab, hw, tn, ot, dm, mf, dv, kw, fx, dq, nm, kf, ox, px, rx, ui, sy])

Medline: ("Deterioration/" or "hospital mortality/" or "vital signs/" or ((clinical adj3 deteriorat*) or "hospital mortality" or "vital sign*" or "emergency department").mp. [mp=ti, ab, hw, tn, ot, dm, mf, dv, kw, fx, dq, nm, kf, ox, px, rx, ui, sy]) and ("artificial intelligence/" or "machine learning" or "artificial neural network/" or "Bayesian learning/" or "classification algorithm/" or "deep learning/" or "fuzzy system/" or "hidden markov model/" or "learning algorithm/" or "supervised machine learning/" or "support vector machine/" or "unsupervised machine learning/" or ("artificial intelligenc*" or "machine learn*" or "artificial neural network*" or "Bayesian learn*" or "classification algorithm*" or "deep learn*" or "fuzzy system*" or "hidden makov model*" or "learning algorithm*" or "supervised machine learn*" or "support vector machin*" or "unsupervised machine learn*").mp. [mp=ti, ab, hw, tn, ot, dm, mf, dv, kw, fx, dq, nm, kf, ox, px, rx, ui, sy])

Filters:

- Publication year: January 2010 – July 2021
- Language: English
- Article type: Journal articles

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Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Machine Learning Techniques for Mortality Prediction in Emergency Departments: A Systematic Review	Page 1 Line 1- 2
ABSTRACT			
Abstract	2	**Objectives:** This systematic review aimed to assess the performance and clinical feasibility of ML algorithms in prediction of in-hospital mortality for medical patients using vital signs at emergency departments. **Design:** A systematic review was performed. **Setting:** The databases including Medline (PubMed), Scopus, and Embase (Ovid) were searched between 2010 and 2021, to extract published articles in English, describing ML-based models utilizing vital signs variables to predict in-hospital mortality for patients admitted at emergency departments. CHARMS checklist was used for study planning and data extraction. The risk of bias for included papers was assessed using the PROBAST tool. **Participants:** Admitted patients to the ED **Main outcome measure:** In-hospital mortality. **Results:** Fifteen articles were included in the final review. We found that eight models including logistic regression, decision tree, K-nearest neighbors, support vector machine, gradient boosting, random forest, artificial neural networks, and deep neural networks have been applied in this domain. Most studies failed to report essential main analysis steps such as data preprocessing and handling missing values. Fourteen included studies had a high risk of bias in the statistical analysis part, which could lead to poor performance in practice. Although the main aim of all studies was developing a predictive model for mortality, nine articles did not provide a time horizon for the prediction. **Conclusion:** This review provided an updated overview of the state-of-the-art and revealed research gaps; based on these, we provide eight recommendations for future studies to make the use of ML more feasible in practice. By following these recommendations, we expect to see more robust ML models applied in the future to help clinicians identify patient deterioration earlier.	Page 1 Line 12- 31
INTRODUCTION			
Rationale	3	The improved life expectancy in Europe and most developed countries has increased admission to the Emergency Departments (ED)s, especially for elderly people who have a higher risk of morbidity and mortality. Also, having more inter- and intra-city travels and organized emergency medical services to transfer these patients to ED brought more importance as well as patient load to EDs. The average rate of mortality among in-patient admissions is around 2% in United States, indicating the necessity of using clinical deterioration prediction models. Moreover, a recent study has shown that some in-hospital deaths due to clinical deterioration can be avoided. Vital signs including heart rate (HR), pulse rate (PR), respiratory rate (RR), oxygen saturation (SpO2), blood pressure (BP), body temperature (TP), and Glasgow coma scale (GCS) are essential parts of the patients' monitoring process. The data received from these parameters could be considered as the cheapest and the most important information, which could easily be collected during admission. While vital signs are among the first data that clinicians observed to have an initial assessment about the patients' condition and have been used in clinical practice for over a century, few studies have effectively quantified their performance in various clinical applications. In recent years, various studies have emphasized that changes in vital signs happen several hours before a serious adverse event. Nowadays, vital signs play a key role in identification of patients with risk of deterioration in ED, however deterioration often occurs unnoticed or is not identified until it becomes too late to intervene. Machine Learning (ML) addresses the question of how to build a computer model that automatically enhances performance through experiments, lying at the intersection of computer science, statistics, and a variety of other disciplines concerned about automatic improvement over time, and inferring and decision making under uncertainty. Various ML algorithms have been developed to solve a wide variety of problems. The most commonly used ML algorithms are supervised learning algorithms focusing on relationships and dependencies between the input and output features. Hence, we can identify the output values for unseen data based on the relationships learned from the training datasets.	Page 2 (line 11-29), Page 3 (line 1-19)

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Section and Topic	Item #	Checklist item	Location where item is reported
		Unsupervised learning is subset of ML that is used in pattern recognition and descriptive modeling without the use of output categorizes or data labeling. Because of enormous advancements in the development of modern ML algorithms such as deep learning, the availability of large databases, and increased computational power, ML has progressed rapidly over the last two decades from laboratory curiosity to practical technology in a wide range of commercial applications. Consequently, utilization of ML techniques in the healthcare domain is increasing rapidly, and the list of tasks where ML has matched or outperformed physicians is growing. Clinical outcome prediction is one challenge receiving attention, and by improvement of ML techniques, the ML-based systems aim to outperform conventional clinical scoring systems. Various studies have been made in this area evaluating the prediction accuracy of in-hospital mortality, cardiac arrest, and intensive care unit (ICU) transfer. To reach this goal, various techniques such as logistic regression (LR), support vector machine (SVM), K-nearest neighbors (KNN), decision tree (DT), random forest (RF), Gaussian process (GP), and artificial neural networks (NN) have been applied, and the results confirmed the appropriateness of using ML in this area. The capability of handling large datasets is the most considerable advantage of ML algorithms, which can result in more accurate prediction of clinical outcomes. Currently, we have various prediction models, using different techniques, in order to improve the triage process and to provide better identification of high-risk patients. Although several studies based on different models were conducted, no single model has yet proved to be superior regarding clinical efficacy.	
Objectives	4	Despite the use of vital signs in patient monitoring and clinical decision making, the importance of specific types of vital signs, their correlation, and frequency of registration to best prevent adverse events and in-hospital mortality is still unclear. However, ML could be a good solution for these challenges. Considering the possible role of ML models in prediction of clinical deterioration of patients in EDs,it is essential to analyze and merge the results of previous studies to have an updated overview of the state-of-the-art. This systematic review focused on the clinical efficacy and technical implications of ML-based models in prediction of mortality using vital signs predictors in patients admitted to EDs.	Page 3 Line 20- 26
METHODS			
Eligibility criteria	5	**Inclusion Criteria:** 1 - Prediction of in-hospital mortality prediction should be the main aim of the study. 2- The study should be done at ED. 3- The study should be done on adult patients. 4- ML algorithms should be used for the prediction task. 5- Vital signs variables should be among the predictors used to build ML models. **Exclusion Criteria**: 1- Studies which are conference articles, posters, abstracts, books or book chapters, and review articles. 2- Studies which were done in pediatrics field. 3- Non-English studies	Page 4, line 7-9 (table 3)
Information sources	6	Medline (PubMed), Scopus, and Embase (Ovid)	Page 4 Line 2
Search strategy	7	**Scopus:** (TITLE-ABS-KEY ((clinical W/2 deteriorat*) OR "mortalit*" or "death" or "hospital mortality" or "vital sign*" or "emergency department") AND TITLE-ABS-KEY ("artificial intelligenc*" or "machine learn*" or "artificial neural network*" or "bayesian learn*" or "classification algorithm*" or "deep learn*" or "fuzzy system*" or "hidden markov model*" or "learning algorithm*" or "supervised machine learn*" or "support vector machine*" or "unsupervised machine learn*")) **Embase**: ("Deterioration/" or "hospital mortality/" or "vital signs/" or ((clinical adj3 deteriorat*) or "hospital mortality" or "vital sign*").mp. [mp=ti, ab, hw, tn, ot, dm, mf, dv, kw, fx, dq, nm, kf, ox, px, rx, ui, sy]) and ("artificial intelligence/" or "machine learning" or "artificial neural network/" or "Bayesian learning/" or "classification algorithm/" or "deep learning/" or "fuzzy system/" or "hidden markov model/" or "learning algorithm/" or "supervised machine learning/" or "support vector machine/" or "unsupervised machine learning/" or ("artificial intelligenc*" or "machine learn*" or "artificial neural network*" or "Bayesian learn*" or "classification algorithm*" or "deep learn*" or "fuzzy system*" or "hidden makov model*" or "learning algorithm*" or "supervised machine learn*" or "support vector machin*" or "unsupervised machine learn*").mp. [mp=ti, ab, hw, tn, ot, dm, mf, dv, kw, fx, dq, nm, kf, ox, px, rx, ui, sy]) **Medline**: ("Deterioration/" or "hospital mortality/" or "vital signs/" or ((clinical adj3 deteriorat*) or "hospital mortality" or "vital sign*" or "emergency	Page 4 Line 6-7 (table 2)

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Section and Topic	Item #	Checklist item	Location where item is reported
		department").mp. [mp=ti, ab, hw, tn, ot, dm, mf, dv, kw, fx, dq, nm, kf, ox, px, rx, ui, sy]) and ("artificial intelligence/" or "machine learning" or "Bayesian learning/" or "classification algorithm/" or "deep learning/" or "fuzzy system/" or "hidden markov model/" or "learning algorithm/" or "supervised machine learning/" or "support vector machine/" or "unsupervised machine learning/" or ("artificial intelligenc*" or "machine learn*" or "artificial neural network*" or "Bayesian learn*" or "classification algorithm*" or "deep learn*" or "fuzzy system*" or "hidden makov model*" or "learning algorithm*" or "supervised machine learn*" or "support vector machin*" or "unsupervised machine learn*").mp. [mp=ti, ab, hw, tn, ot, dm, mf, dv, kw, fx, dq, nm, kf, ox, px, rx, ui, sy])	
Selection process	8	Three databases including Medline (PubMed), Scopus, and Embase (Ovid) were targeted. Relevant articles were extracted using a broad range of relevant keywords. We stratified keywords into five groups namely, ML keywords, medical keywords, document type, publication year, and language. The keywords in a group were paired using OR operators and all groups were paired using AND operator. Table 2 shows the search keywords that were applied to the titles, abstracts, and full text in the three databases	Page 4 Line 1-6
Data collection process	9	For item extraction were prepared based on the critical appraisal and data extraction for systematic reviews of prediction modelling studies (CHARMS) checklist. The extracted items include study design, publication date, source of data, study population, outcomes, time horizon, list of vital signs predictors, data preprocessing, model development, model performance, model validation and evaluation, and interpretation.	Page 4 Line 10-16
Data items	10a	Mortality is the outcome.	Page 4 Line 0 (Table 1)
	10b	Study year, country, study type (retrospective, prospective), population, outcome portion (%), patients type, mean age, male (%), vital signs, machine learning algorithms, evaluation metrics, time horizon for prediction, personalized model, handling missing values, hyperparameter optimization, approach for prediction, validation	Page 6, line 6 (table 4) Page 8, line 1 (table 5)
Study risk of bias assessment	11	Risk of bias (ROB) for each study was assessed by using the prediction risk of bias assessment tool (PROBAST) checklist and reported based on an adapted form. Extracted articles were classified into three different categories (low, high, unclear). A study is considered as having a high ROB if the study has at least one high ROB in four domains. ROB assessment of each study was checked by two of researchers (AN, MNB); *disagreements were resolved in collaboration with a senior researcher* (UKW).	Page 5 Line 1-6
Effect measures	12	It is nor relevant to this study.	-
Synthesis methods	13a	Three researchers (AN, MM, AE) screened the title and extracted abstracts, independently. The screening process was done using the Covidence tool. Then, two researchers (AN, TS) read the full texts that were extracted after the title and abstract screening phase to retrieve the final set of articles that met our objectives and inclusion criteria.	Page 4 Line 9-16
	13b	-	
	13c	PRISMA flowchart was used to visualize the process of selecting the final set of studies. Also extracted information of studies are presented by two tables (table 4 and table 5).	Figure 1, Page 6, line 6 (table 4) Page 8, line

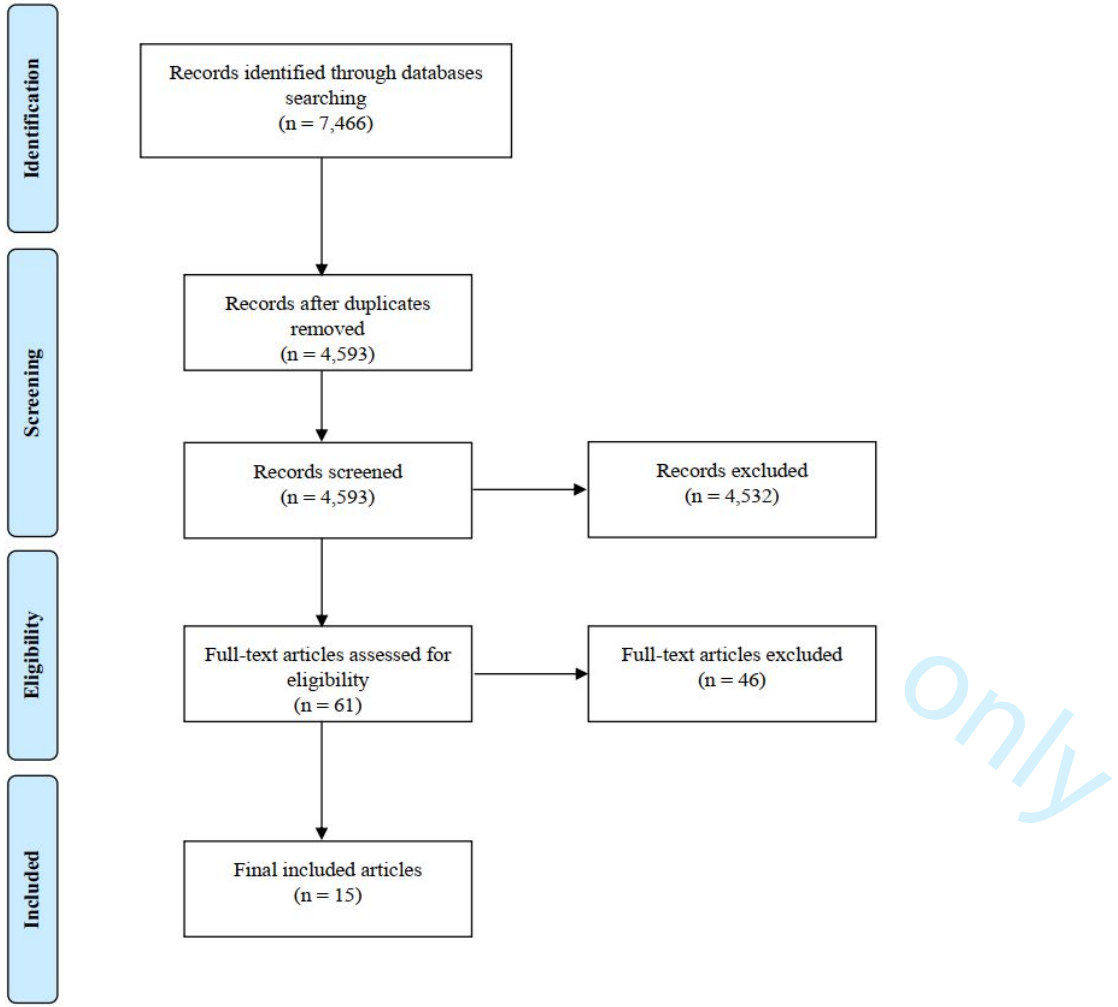
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Section and Topic	Item #	Checklist item	Location where item is reported
			1 (table 5)
	13d	-	
	13e	-	
	13f	-	
Reporting bias assessment	14	(PROBAST) checklist is used in this study.	Page 11, line 1 (table 7)
Certainty assessment	15	The outcome is obvious (in-hospital mortality).	Page 4 Line 0 (Table 1)
RESULTS			

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Section and Topic	Item #	Checklist item	Location where item is reported
Study selection	16a		Figure 1
	16b	-	

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Section and Topic	Item #	Checklist item	Location where item is reported
Study characteristics	17		Page 6, line 6 (table 4) Page 8, line 1 (table 5)

Table 4. General characteristics of included studies

ID	Authors	Year	Country	Study type	Population	Outcome portion (%)	Patients type	Mean age (y)	Male (%)	Vital signs
A1[21]	Faisal et al.	2020	UK	Retrospective	24,696	5.33	All admissions	63	46.9	RR, SpO2, SBP, PR, GCS, TP
A2[29]	Raita et al	2019	USA	Retrospective	135,470	2.10	All admissions	46	56.8	TP, PR, SBP, DBP, RR, SpO2
A3[19]	Klug et al.	2020	Israel	Retrospective	799,522	1.55	All admissions	55	51.5	TP, HR, SBP, DBP, SpO2
A4[22]	Kwon et al.	2020	South Korea	Retrospective	23,587	3.98	Infectious patients	63	46.1	SBP, RR, TP, HR
A5[31]	Liu et al.	2011	Singapore	Retrospective	100	40.00	All admissions	65	63	RR, PR, SBP, DBP, SpO2, GCS
A6[44]	Taylor et al.	2015	USA	Retrospective	5,278	4.92	Infectious patients	65	55	HR, TP, SBP, DBP, RR, SpO2
A7[18]	Levin et al.	2017	USA	Retrospective	172,726	0.40	All admissions	46	44.7	TP, HR, RR, SBP, SpO2
A8[43]	Perng et al.	2019	Taiwan	Retrospective	42,220,	4.71	Infectious patients	64	56.5	SBP, GCS, TP, HR
A9[39]	Shamout et al.	2020	UK	Retrospective	37,284	0.80	All admissions	68	48.8	HR, SBP, RR, TP, SpO2
A10[33]	Chen et al.	2021	Taiwan	Retrospective	52,626	9.4	Infectious patients	72	58.3	HR, SBP, DBP, TP, RR, SpO2, GCS
A11[34]	Joseph et al.	2020	US	Retrospective	445,925	< 13	All admissions	53	45.9	HR, SBP, DBP, RR, TP, SpO2
A12[32]	Karlsson et al.	2021	Sweden	Retrospective	445	14.1	Infectious patients	73	52.6	HR, SBP, DBP, RR, TP, SpO2
A13[35]	Rodriguez et al.	2021	Columbia	Retrospective/ Prospective	2,510	11.5	Infectious patients	62	49.8	SBP, DBP, TP, RR, SpO2, GCS
A14[36]	Soffer et al.	2020	Israel	Retrospective	118,262	5.3	All admissions	73	52.6	HR, SBP, DBP, TP, RR, SpO2
A15[37]	van Doorn et	2021	Netherlands	Retrospective	1,344	15	Infectious	71	54.4	HR, SBP,

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Section and Topic	Item #	Checklist item	Location where item is reported

	al.						patients			DBP, TP, RR, SpO2, GCS

HR: heart rate; PR: pulse rate; RR: respiratory rate; SpO2: oxygen saturation; SBP: systolic blood pressure; DBP: diastolic blood pressure; TP: body temperature: GCS: Glasgow coma scale

Table 5. Selected articles list and their ML-related characteristics

Id	ML algorithms	Evaluation Metrics	Time Horizon	Personalized	Handling missing values	Hyperparameter Optimization	Approach	Validation
A1	LR, RF, SVM, NN	AUC, Brier Score	-	No	No	Yes	Binary classification	External
A2	LR, RF, GB DT, DNN	AUC, Sensitivity, Specificity	-	No	No	Yes	Binary classification	Internal
A3	GB	AUC, Sensitivity, Specificity, Brier Score	-	No	No	No	Binary classification	Internal
A4	GB, RF	AUC	72h	No	No	Yes	Binary classification	External
A5	NN, SVM	Accuracy, Sensitivity, Specificity	9-12 minutes	No	No	No	Binary classification	Internal
A6	RF, DT, LR	AUC	-	No	Yes	No	Binary classification	Internal
A7	RF	AUC	-	No	No	No	Binary classification	Internal
A8	KNN, SVM, RF, DNN	AUC, Accuracy	72h, 28 days	No	Yes	No	Binary classification	Internal
A9	DNN	AUC	2h, 24h	Yes	Yes	No	Time series regression	External
A10	SVM, GB, NN	AUC, accuracy, Sensitivity, Specificity, PPV, NPV	-	No	Yes	No	Binary classification	Internal
A11	LR, DNN, GB	AUC, accuracy, Sensitivity, Specificity,	-	No	No	Yes	Binary classification	Internal
A12	RF	AUC, Sensitivity, Specificity, PPV, NPV, PLR, NLR	7days, 30 days	No	No	No	Binary classification	Internal
A13	DT, RF, NN, SVM	AUC, accuracy	-	No	No	No	Binary classification	Internal
A14	GB	AUC, Sensitivity, Specificity, NPV, PPV, FPR	-	No	Yes	No	Binary classification	Internal
A15	LR, NN,	AUC, Sensitivity,	31 days	No	No	No	Binary	Internal

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Section and Topic	Item #	Checklist item	Location where item is reported
		RF, GB | Specificity | | | | | classification |	
		ML: machine learning, LR: logistic regression, SVM: support vector machine, KNN: K-Nearest neighbors, DT: decision tree, RF: random forest, GP: gaussian process, NN: neural networks, DNN: deep neural networks, AUC: area under the curve, PPV: positive predictive value, NPV: negative predictive value, PLR: positive likelihood ratio, NLR: negative likelihood ratio, FPR: false positive rate.	
Risk of bias in studies	18	**Table 7. Risk of bias (ROB) assessment of included studies according to PROBAST checklist**	Page 11, line 1 (table 7)
Results of individual studies	19	There is only one outcome: in-hospital mortality.	Page 4 Line 0 (Table 1)
Results of syntheses	20a	-	
	20b	-	
	20c	-	
	20d	-	
Reporting biases	21	-	
Certainty of evidence	22	-	
DISCUSSION			

ID	PROBAST items				
	Participants	Predictors	Outcomes	Sample size & missing data	Statistical analysis
A1	+	+	+	-	-
A2	+	+	+	-	-
A3	+	?	+	-	-
A4	+	?	+	-	-
A5	-	?	+	-	-
A6	+	+	?	+	?
A7	+	+	+	-	-
A8	+	?	?	-	?
A9	+	+	+	+	?
A10	+	+	+	+	-
A11	+	+	+	?	+
A12	-	+	+	-	-
A13	?	+	+	-	?
A14	+	+	+	?	?
A15	?	+	+	-	?

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Section and Topic	Item #	Checklist item	Location where item is reported
Discussion	23a	This review consisted of fifteen studies that developed ML-based models to predict mortality in EDs. Our analysis showed that most studies described the datasets in sufficient detail and had enough number of participants. Moreover, the outcome domain had the lowest risk of bias which shows most studies had a clear outcome definition. However, we found several methodological and reporting shortcomings, and this can lead to poorer performance than reported when the models are deployed in real life. Many studies skipped or failed to report important phases such as data preprocessing, handling missing values, and model development. It has been stated that up to 80% of data analysis is spent on data cleaning and data preparation, so these phases are very crucial for having a ML predictive model with good performance. It also could be one of the reasons why some researchers have discussed the poor effectiveness of ML predictive models in clinical environments. Moreover, despite the dramatic advances in ML and introducing novel and effective models, various researchers still have applied traditional ML techniques such as LR and DT. For instance, we found that LR is the third most common algorithm used in the studies. Most ML models have parameters that needed to be tuned to gain the best performance and generalization power and avoiding overfitting, but most studies did not provide information about the models' validation process and finding the hyperparameters of the models. Of fifteen studies, fourteen studies are retrospective studies and only three studies have evaluated their models based on external validation. The ultimate aim of included studies was to predict mortality in EDs ahead of time to provide enough time for clinicians to intervene and prevent mortality. However, only six studies have provided their time horizon, the time that clinicians have to treat the patients. The other studies have built binary classification algorithms to stratify patients into two classes, and their models do not have the capability to be used in practice.	Page11 (line 4-16) - page 12 (1-6)
	23b	-	
	23c	To the best of our knowledge, this study is the most recent systematic review on the topic of using ML algorithms to predict mortality at emergency departments. This study was done with the help of an experienced artificial intelligence expert, an experienced research librarian who is familiar with the health research area, and three expert clinicians at ED. Moreover, ROB analysis was done using PROBAST checklist which has four domains that assess different aspects of studies. Data extraction was done using a standard checklist called CHARMS. However, our study has some limitations. First, it was not possible to quantify the analysis or a conduct meta-analysis due to the high heterogeneity of included studies. Also, we have considered the vital signs due to high availability at ED, while the role of other clinical predictors in patients' clinical conditions should be investigated.	Page 14 (Line 21-27)- page 15 (line 1-2)
	23d	Perhaps in this research area, the main challenge is integration of ML models into clinical practice. A recent systematic review based on performance metrics, particularly AUC, showed that ML predictive models outperformed usual care in most detection and prediction tasks at ED. However, the authors mentioned that many studies have limited applicability to clinical practice and there are other considerations more than performance metrics as well as barriers that should be taken into account to have successful ML models in real life. Therefore, despite the great research successes in building ML-based predictive models for clinical practice, there remain few examples of ML models being successfully integrated into the daily routine or critical parts of clinical environments. This reveals that what is being done in research is not completely in line with the realities of clinical practice. In this section, we mention some of these issues and technical barriers, then present recommendations on how to address the most prominent issues, and encourage researchers to take them into account. Our analysis indicated that most studies are retrospective. However, the impact of ML techniques in clinical environment will need further validation in randomized control trials and prospective studies before widespread clinical adoption where the data could be incomplete and noisy with high level of uncertainty. Recommendation 1: we recommend that besides building models using historical collected data, researchers consider prospective analysis and think about building adaptive models that can monitor patients in real-time. Although external validation is one of the most rigorous ways to assess the generalization power of predictive models, it is still one of the common approaches for showing the robustness of predictive models. Recommendation 2: we recommend that researchers investigate the performance of their models using external validation to have a better estimation of the generalization power of their models. Recommendation 3: regarding the preprocessing phase, since some steps such as data cleaning, handling missing and noisy values are an inevitable part of each data analysis study, we recommend that researchers instead of ignoring records with missing values, use proper techniques to handle noisy and incomplete records. For instance, we recommend that complete case analysis is avoided and proper imputation techniques such as iterative multivariate imputation technique which is one of the best approaches for imputing missing values is used instead. Another challenge in most studies is that the number of event patients is too low compared to non-event patients. This can lead to bias and overfitting during developing ML	Page 12 (line 20-28), page 13 (1-28), page 14(1-20)

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
		models, so proper under/oversampling techniques such as SMOTE and its variations should be utilized to make a balance between classes before developing models. Another issue is time horizon for prediction. Gerry et al. recommended that time horizon should be limited to a few days at most, since signs of deterioration will probably not be observed for more than a few days. Recommendation 4: we suggest that researchers in this area choose a realistic time horizon rather than prediction of mortality after six or twelve months. Moreover, most of the included studies are based on building a single model for the whole population. Recommendation 5: as patients have different characteristics and norms, we recommend that patients' characteristics, vital signs trajectories, trends, etc. are used for each individual patient to tune ML models to have a better performance in practice. Recommendation 6: we recommend that researchers find a way to make their results and workflow understandable and interpretable by humans. For instance, deep learning is one of the hottest research topics today, and it has been applied in different applications of clinical practice, but it has inherent limitations. For example, despite impressive results, it is not indicative of high-level reasoning and rationality. Such methods are often considered as a "black box", where the decision logic is presented in millions of numerical weights and biases. This lack of transparency could have some legal and ethical implications and might increase distrust of ML models by patients and clinicians, and it might be one of the reasons that patients are likely to trust a clinician more than a machine. This has been described as a crucial problem for ML acceptance in clinical practice and attempts are underway to create more human interpretable models. For example, with the help of image processing techniques, it is possible now to visualize the input and output of ML models; this makes the understanding of ML models easier. Lack of widely accepted and utilized reporting or publication guidelines for implementing ML in clinical practice is a challenge that brings difficulty to the research quality assessment, especially for clinicians without a strong mathematics and computer science background. Another challenge for ML models is that they need a huge amount of labeled data to be trained, and the performance of ML models often depends on the performance of the human labeling the data. Recommendation 7: as this area is interdisciplinary in nature, it is imperative that clinicians and artificial intelligence experts become better at interdisciplinary collaboration to become more familiar with the needs, limitations, and challenges in clinical and technical domains, which ultimately increases the quality of the predictive models. Recommendation 8: we recommend that researchers should be careful and accurate when reporting their studies. It has been shown that the quality of reporting of predictive models in this clinical domain is poor. Only with full and transparent reporting of information about all aspects of a predictive model, the risk of bias and the usefulness of that model can be assessed. It is recommended that researchers use available checklists such as SPIRIT-AI for designing the trial, and TRIPOD and CONSORT-AI for reporting the findings to make sure that they provide sufficient information to clearly and accurately report their studies.	
OTHER INFORMATION			
Registration and protocol	24a	-	
	24b	-	
	24c	-	
Support	25	There is no funding attached to this systematic review.	Page 15 Line 22
Competing interests	26	None declared.	Page 15 Line 23
Availability of data, code and other materials	27	All data relevant to the study are included in the article or uploaded as online supplementary information.	Page 15 (Line 25-26)

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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